

An Obstruction Calculus for Viability under Incomplete Observation

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Abstract

Under perfect measurement the viability kernel — the set of states from which some feedback keeps the system within its constraints — is characterized by tangency conditions. Under incomplete observation the sufficiency direction has a canonical answer in Veliov’s output-feedback regulation condition and the estimation-tube reduction. The complementary direction — certifying that no observation-based policy is viable — has received less systematic treatment.

This paper develops that instrument: a calculus of obstruction certificates, sound sufficient conditions for nonviability (equivalently, necessary conditions for viability), finitely checkable in the common-action, fibre-certification, and finite-horizon forms and analytic drift-and-timing conditions elsewhere. They do not exhaust the complement of the epistemic kernel, the kernel’s observation-based counterpart; in finite systems backward recursion is complete. Five mechanisms are established — two finitely checkable, two closed-form conditional, one minimal — with a sixth under a policy-class restriction. The central common-action obstruction shows that when the safe controls of compatible states intersect emptily, no observation-based policy is viable, though every compatible state is individually viable under full information. The others are a finite-time exit certificate under an Isaacs-type drift condition, an epistemic-emptiness construction in which a constant observation merges states with incompatible admissible controls, a delayed-information obstruction with an explicit timing bound, and a certification limit: an exact observation-only certifier exists iff safe-set membership is constant on the observation fibres. The sixth, a certainty-equivalence trap, empties the kernel of the uncorrected controller under a biased observation even when the perfect-information kernel is nonempty. Two certification-theoretic completions are established: over polytope-declared blind-window classes the timing certificate is exactly computable as a linear program, with the Farkas dual as the obstruction witness, sound beyond the declaration with an unbounded implementable-class gap; and the residual dynamic gap is shown to decompose exactly, its post-observation recourse phase undecidable by any certificate pair that reads only the window sub-model.

The calculus is positioned against barrier certificates and estimation tubes, and its monitoring and institutional consequences — observation timing, coarseness, aggregation, and bias — are drawn.

Keywords: viability kernel; obstruction certificates; robust viability; epistemic viability; output feedback; sustainability governance; partial observation

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1. Introduction

1.1 The problem

Viability theory characterizes the states of a constrained control system from which at least one control can keep every future state within a constraint set (Aubin, 1991). For sustainability problems the constraints are floors — stock levels, service thresholds, safety margins — and viability is the formal counterpart of maintaining a development path rather than merely optimizing it (Béné, Doyen, and Gabay, 2001; De Lara and Doyen, 2008; Doyen et al., 2012; Doyen and Gajardo, 2020). The theory under *perfect measurement* is complete in the relevant sense: the viability kernel exists as the largest viable subset, and Nagumo-type tangency conditions certify it.

Sustainability governance, however, operates under *incomplete observation*. Stocks are assessed at discrete review intervals. Indicators are coarser than the state. Some stocks are unobserved altogether. And the information that does arrive may arrive late. This paper addresses that side: for an incomplete observation structure, it develops instruments that certify that no observation-based policy is viable — that the infeasibility is due to the information structure, not to the dynamics.

Floors in sustainability assessment are typically constraints on the productive base — the state of the system that yields the measured output — rather than on the output itself. Such a base can be read as natural capital, as a stock, or as a slowly regenerating flow of services; these are overlapping readings of the same asset rather than mutually exclusive ones, and what makes a use of it sustainable is whether the use falls on the yield or on the base itself. Across assets, whether the base behaves as a flow or as a stock is a continuum, set by its regeneration timescale relative to the rate of use — a continuum that runs from a season, through a year, to geological time for a mineral deposit. If a base regenerates too slowly for the rate at which it is taken, drawdown is still liquidation.

A base can also erode while measured output is maintained, because use draws on the base rather than on its yield: the measured quantity can be held up while the base declines, and partial observation may not detect the

erosion until a floor is breached. The signal that would reveal the erosion is accordingly often the one that is coarse, postponed, or absent. Incomplete observation is therefore structural to the certification problem, not merely a practical limitation; the sufficiency literature and the obstruction calculus of this paper address the same underlying indeterminacy rather than competing ones.

The question splits naturally. The sufficiency direction has a canonical literature: Veliov (1993) gives conditions for the existence of an output-feedback regulation map under incomplete and inexact measurement, reducing under perfect measurement to the classical viability condition (Haddad, 1981); the estimation-tube programme (Cardaliaguet, Quincampoix, and Saint-Pierre, 2007) absorbs imperfect information into an estimation space of sets of compatible states with no loss of value; and barrier certificates certify safety (Prajna and Jadbabaie, 2004; Prajna, Jadbabaie, and Pappas, 2007), with converse and hybrid necessary-and-sufficient characterizations (Prajna and Rantzer, 2005; Maghenem and Sanfelice, 2019). What that literature does not supply is the converse direction: a calculus of obstruction certificates, each a sufficient condition for nonviability whose negation is a necessary condition for observation-based viability. An obstruction certificate is a checkable witness that no observation-based policy exists — finite in the finite-state and polyhedral cases — and the argument it makes is that a prescribed class of observation-based policies fails not because a particular policy is bad, but because the information structure leaves no room for any policy. In the polyhedral common-action and certification forms (Theorem 2 and Proposition 7) it is a finite, checkable test; the drift certificates of Theorem 5 and Theorem 3 are not finite objects, one exhibiting an enforcement selection and the other bounding the timing uniformly over every policy. Veliov’s condition tells us when output feedback can work; the obstruction calculus tells us when it cannot, and for the borderline cases it identifies which quantitative feature of the observation design — timing, coarseness, bias, aggregation — is responsible.

1.2 Contributions

Five obstruction mechanisms are developed, each with a complete proof, and a sixth is exhibited under a policy-class restriction:

1. **The instantaneous common-action obstruction** (*finitely checkable*) (Theorem 2, Example 1). If the safe-control sets of the states compatible with the current information set intersect emptily, then no observation-based policy is viable, even though every compatible state is individually viable under full information. The failure is purely informational: no stochasticity, no estimation error.
2. **The delayed-information obstruction** (*closed-form conditional*) (Theorem 3). If the information needed to distinguish safe from unsafe compatible states arrives only at time T_{obs} , and the disturbance can force constraint violation earlier than T_{obs} , then no observation-based policy is viable. The timing bound $\inf q/\varepsilon < T_{\text{obs}}$ is stated explicitly: information may be accurate but arrive too late.
3. **The finite-time exit certificate** (*closed-form conditional*) (Theorem 5). Under a Dini-drift condition of Isaacs type — the disturbance chooses after the control — the disturbance can force exit from the safe set within an explicit time bound, for *every* admissible control of any information structure. This is the base obstruction: when it applies, the viability question is closed without any observation-theoretic argument.
4. **Epistemic emptiness by admissibility** (*minimal construction*) (Proposition 5, a minimal construction). A constant observation merges states whose *admissible* controls differ: the state-dependent control set forces a single common action, and that action is unsafe at the boundary, so the unresolved initial observation fibre lies outside the epistemic kernel, even though every physical state is viable under full information. The mechanism is admissibility — the common-action obstruction of Theorem 2 at minimal size — and Example 1 gives the hidden-mode instance.
5. **The fibre certification criterion** (*finitely checkable*) (Proposition 7, Corollary 1). An exact observation-only safety certifier — a function of the observation that returns “safe” exactly on the safe set — exists if and only if safe-set membership is constant on every observation fibre. The certainly-safe set is the sound relaxation: the largest set of observations that can be labeled safe without further information.
6. **The certainty-equivalence trap** (*policy-class restriction*) (Remark 5). Even an *injective* observation empties the kernel of a system whose perfect-information kernel is nonempty for the canonical certainty-equivalence controller — the perfect-information zeroing law applied to the uncorrected observation. The trap is a policy-specific failure: a controller that uses the observation map’s policy-specific failure: a controller that uses the observation map’s structure (the bias correction) restores viability.
7. **Computable instantiation and the exact residual** (*finitely checkable; exact residual*) (Theorem 4, Example 2, Propositions 9 and 10). Over polytope-declared blind-window classes the timing certificate is exactly computable as a linear program — polynomial size, exact in rational arithmetic, Farkas-certified on failure — sound beyond the declaration, with the implementable-class gap shown unbounded by an explicit instance on which the relaxation extinguishes the certificate. The residual gap of the completeness question decomposes exactly into the window phase, decided completely by the certificates, and the post-observation recourse phase, decided by no window-measurable certificate pair.

The calculus is organized by a single selector principle (Proposition 1, Section 2.4): observation-based viability is the existence of one response — admissible, safe, and recursively viable — for every compatible state, and each certificate proves that a specific response correspondence

is empty; the obstructions are nested in a ladder (Proposition 3). Five refinements complete the picture: backward belief recursion is sound and complete on any finite horizon in finite systems (Theorem 1); a uniform-margin condition connects the instantaneous and tube obstructions (Proposition 2); the timing obstruction has a sharp threshold form (Remark 2); a Helly-type sparse witness bounds the common-action obstruction by $m + 1$ compatible states (Proposition 4); and a comparison-function form sharpens the exit-time bound (Remark 4). The epistemic kernel is monotone in action sets, disturbances, information, and policy class (Proposition 8).

Throughout, “obstruction” means a *certified* failure. Theorem 5 and Theorem 3 each exhibit the violating constraint, an admissible disturbance, and the quantitative bound that make failure inevitable, for every policy in the prescribed class; Proposition 5 and Theorem 2 exhibit the incompatible admissible controls — the merged observation whose incompatible safe controls certify emptiness — with no quantitative bound claimed there; and Proposition 7 is the converse-side characterization of the certification limit itself. The calculus is positioned against the barrier-certificate literature (Section 6.2) and the estimation-tube programme (Section 6.3), and its consequences for the design of monitoring and indicators in sustainability governance are drawn in Section 6.4.

1.3 Related work

The viability-theory background is Aubin (1991), Aubin, Bayen, and Saint-Pierre (2011), and the kernel approximation theory of Saint-Pierre (1994); robust viability is Aubin and Frankowska (1990) and Frankowska (1989). Under incomplete measurement, the sufficiency theorem is Veliov (1993), viability with a priori unknown but observable parameters is Quincampoix and Veliov (1994), and the estimation-set reduction is Cardaliaguet, Quincampoix, and Saint-Pierre (2007). On the failure side, Aubin (2001) characterizes the complement of the kernel — Poincaré’s “shadow” — together with the capture basins, and Aubin and Catté (2002) supply the bilateral-fixed-point and discriminating-kernel calculus; the certificates of Section 3 are one-sided, observation-constrained analogues of that programme. In verification, barrier certificates originate with Prajna and Jadbabaie (2004), with worst-case and converse results in Prajna, Jadbabaie, and Pappas (2007) and Prajna and Rantzer (2005) and hybrid characterizations in Maghenem and Sanfelice (2019). The sustainability application domain is Béné, Doyen, and Gabay (2001), De Lara and Doyen (2008), Doyen et al. (2012), and Doyen and Gajardo (2020). To our knowledge, the obstruction certificates of Theorem 2, Theorem 3, and Proposition 7 — in particular the certification criterion and the timing bound — have not been stated in this form; the elementary facts on which they rest (quantifier commutation and Dini comparison) are classical.

1.4 Organization

Section 2 fixes the framework: the control system, the viability kernel hierarchy, observation structures, information states, the safe-control correspondence, and the selector principle (Proposition 1). Section 3 develops the obstruction calculus (Theorem 1–5, Propositions 2–5, Remarks 2–

4, and Example 1): finite-horizon completeness, the instantaneous common-action obstruction with its ladder and sparse witness, the delayed-information obstruction with its threshold form, and — as supporting certificates — the finite-time exit certificate and the admissibility obstruction, together with a recourse certificate for the post-observation mode (Proposition 6, Example 3). Section 4 develops the certification limits (Proposition 7, Corollary 1, Remark 5). Section 5 reviews the sufficiency landscape, cited rather than re-derived. Section 6 discusses the position of the calculus relative to barrier certificates and estimation tubes, states the monotonicity of the epistemic kernel (Proposition 8), and draws the governance consequences. Sections 7–10 develop the new material: complete characterizations and the residual gap (Section 7), the selector principle with its dual witnesses and the monitoring-adequacy converse (Section 8), a calibrated case study with a coverage audit (Section 9), and the probabilistic belief-state development (Section 10). Section 11 concludes. Brief sketches and short proofs of the main-text statements are given in place; the extended discussions and worked instances of the Sections 3, 4, and 10 certificates, the complete proofs of the Section 3 certificates, the expanded sufficiency review, and the auxiliary constructions are collected in the Supplementary Material; the results of Sections 7 and 9 are proved in the main text. The running instances are fixed here for orientation: the two-floor conflict (Supplementary S1), on which the worked instances of the Sections 3, 4, and 10 certificates are built, and the calibrated delayed-hidden-regime case study (Section 9).

2. Framework

2.1 The control system and the kernel hierarchy

This section fixes the mathematical objects used throughout the paper: the control system, the disturbance structure, the constraint set, and the four kernels (perfect-information, robust, epistemic, and institutionally restricted) that organize the analysis.

Fix a state space $X \subseteq \mathbb{R}^n$, a control set U , a disturbance set D , and dynamics

$$\dot{x}(t) = f(x(t), u(t), d(t)), \quad u(t) \in U(x(t)), \quad d(t) \in D(x(t)),$$

with f continuous, U and D set-valued with closed graph and nonempty compact values, and admissible controls and disturbances taken to be measurable selections — the standing solution-existence conditions: every selection pair admits a Carathéodory solution, and the convexified relaxed inclusion of Section 3.5 satisfies the Marchaud conditions of the viability literature (Aubin, 1991). Let $\mathcal{V} \subseteq X$ be a closed constraint set. A *causal policy* π is a map from the observation record to controls, respecting the specified information structure. The standard objects are:

- $\text{Viab}(\mathcal{V}; U, \pi_{\text{perf}})$: the **viability kernel** under full-information (state-feedback) policies — the largest closed subset of $\mathcal{V}$ from which there exists a state-feedback control keeping the trajectory in $\mathcal{V}$ for all time — no disturbance, or a known disturbance trajectory (Aubin, 1991).
- $\text{RViab}(\mathcal{V})$: the **robust kernel**, the largest subset of $\mathcal{V}$ from which there exists a state-feedback policy keeping every trajectory — for *every* admissible disturbance realization — in $\mathcal{V}$ (Aubin and Frankowska, 1990).

- $\text{ERViab}_{\mathcal{I}}(\mathcal{V})$: the **robust epistemic kernel** — the object of this paper (Definition 1). Its elements are *states of information*, not physical states (Section 2.3). The non-robust counterpart $\text{EViab}_{\mathcal{I}}(\mathcal{V})$ is the non-robust contrast class (Section 2.3).

Projected to physical state space — with $\text{proj}_{\exists}(\mathfrak{B}) = \bigcup_{B \in \mathfrak{B}} B$ the *existential* projection of a collection $\mathfrak{B}$ of information states, applied to the first two terms, whose elements are information states — the informational hierarchy reads

$$\begin{aligned} \text{proj}_{\exists}(\text{IRViab}_{\mathcal{I}}(\mathcal{V})) &\subseteq \text{proj}_{\exists}(\text{ERViab}_{\mathcal{I}}(\mathcal{V})) \\ &\subseteq \text{RViab}(\mathcal{V}) \subseteq \text{Viab}(\mathcal{V}), \end{aligned}$$

where $\text{IRViab}_{\mathcal{I}}(\mathcal{V})$ is the institutionally restricted kernel of Section 6.4. The physical implication of epistemic viability is the inclusion at the level of beliefs: $B_0 \in \text{ERViab}_{\mathcal{I}}(\mathcal{V}) \Rightarrow B_0 \subseteq \text{RViab}(\mathcal{V})$. Each inclusion can be strict, with a distinct cause: robust contraction arises from disturbances; epistemic contraction from indistinguishability; institutional contraction from restricted authority, enforcement, and allocation. The present paper characterizes the *epistemic* contraction: the mechanisms by which indistinguishability empties or shrinks the kernel.

2.2 The safe-control correspondence

For a constraint set $\mathcal{V}$ described by finitely many C^1 constraint functions, $\mathcal{V} = \{x : q_j(x) \geq 0, j = 1..m\}$, define the **safe-control correspondence** at $x \in \mathcal{V}$:

$$\mathcal{R}_{\mathcal{V}}(x) = \{u \in U(x) : \forall d \in D(x), \forall j \text{ with } q_j(x) = 0, \nabla q_j(x) \cdot f(x, u, d) \geq 0\},$$

the set of controls whose drift is inward (subtangential) on every active constraint face, uniformly over the disturbance. For x in the interior of $\mathcal{V}$ (no active constraints), $\mathcal{R}_{\mathcal{V}}(x) = U(x)$. Two levels of the tangency condition are distinguished throughout. (i) *Local reading*. The correspondence is applied to the constraint set itself: by Nagumo's theorem (Nagumo, 1942) in its robust form, if $\mathcal{V}$ is closed and $\mathcal{R}_{\mathcal{V}}(x) \neq \emptyset$ on $\partial\mathcal{V}$ with a measurable selection $u(x) \in \mathcal{R}_{\mathcal{V}}(x)$, then $\mathcal{V}$ is itself robustly invariant, i.e. $\text{RViab}(\mathcal{V}) = \mathcal{V}$ (Aubin, 1991; Frankowska, 1989). (ii) *Kernel reading*. When $\mathcal{V}$ is not invariant, viability of a point $x_0 \in \mathcal{V}$ requires instead a selection $u(x) \in \mathcal{R}_G(x)$ along the trajectory, where $G = \text{RViab}(\mathcal{V})$ and $\mathcal{R}_G$ is the same correspondence computed on the kernel: points of $\mathcal{V} \setminus G$ may admit controls that are instantaneously safe while every trajectory must exit later. The obstruction theorems below are *local certificates against* $\mathcal{R}_{\mathcal{V}}$: exhibiting a state at which even the weaker pointwise condition fails certifies exclusion from the kernel, and this is the reading in which the certificates are used. The correspondence is the bridge between the pointwise (tangency) and the global (kernel) description, and it is the object the observation structure acts upon: an observation-based policy can select controls only as a function of the observation record.

2.3 Observation structures, information states, and epistemic kernels

Conventions. Throughout, the plant evolves in continuous time and a policy acts at review times, holding the selected action until the next review; the observation record is generated by the observation map O (the error, hidden-parameter, and delay extensions are noted below). Between reviews the record $y(t) = O(x(t))$ is observed continuously but cannot change the held action; the certificates of Sections 3.2–3.3 and 3.6 use the record only through what it discriminates (observation-equivalence), so they remain valid under any refinement of the within-interval record that does not separate compatible

branches. A policy is observation-based if its action depends on the record only through the information set B_t ; under the delay-free set-membership semantics used here the information set is a sufficient statistic, so observation-based and record-based policies coincide (with delays, hidden parameters, or history-dependent constraints the belief must be augmented and sufficiency is not automatic; Section 6.5). The disturbance is chosen adversarially after the control (the Isaacs order of Theorems 5 and 2), an action outside $U(x)$ at a compatible state x is itself a failure (Section 2.4), and constraint violation is the strict inequality $q < 0$.

An **observation structure** $\mathcal{I} = (Y, O)$ is a measurable space Y and an observation map $O : X \rightarrow Y$. The policy observes $y(t) = O(x(t))$. At time t , the **information set** (belief) of a regulator who knows the dynamics, the disturbance class, its own applied controls $(u(s))_{s < t}$, and the record $(y(s))_{s \leq t}$ is

$$\begin{aligned} B_t &= \{\xi(t) : \xi(0) \in B_0, \\ &\quad \dot{\xi}(s) = f(\xi(s), u(s), d(s)), d(s) \in D(\xi(s)), \\ &\quad O(\xi(s)) = y(s) \text{ for all observed } s \leq t\}, \end{aligned}$$

where the applied control history $u(\cdot)$ in the dynamics is the regulator's own — common to every candidate trajectory — and $\xi(\cdot)$ ranges over candidate trajectories: the set of current states compatible with the record and the applied controls. We assume the standard set-membership semantics (Kurzhanski and Vályi, 1997), so that B_t contains every state consistent with the observations and the applied controls. (With observation error, the equality becomes membership $y(s) \in H(\xi(s), v(s))$; with hidden parameters the belief lives in $X \times \Theta$; with delays it lives in a history space — the extensions are noted as open in Section 6.5.) A policy is **observation-based** if $u(t)$ depends on the record only through B_t ; this is a special case of record-based policies, and the two coincide when the belief is a sufficient statistic for the decision problem, which holds in the delay-free set-membership semantics used throughout. We write $U^B(B) = \bigcap_{x \in B} U(x)$ for the controls available at every compatible state — the only controls an observation-based policy may select without risking inadmissibility.

Definition 1 (robust epistemic kernel). $B_0 \in \text{ERViab}_{\mathcal{I}}(\mathcal{V})$ if there exists a nonanticipative observation-based policy π such that, for every compatible initial state $x_0 \in B_0$ and every admissible disturbance realization $d(\cdot)$, the trajectory of $\dot{x} = f(x, u(t), d(t))$ generated from x_0 by π 's actions under $d(\cdot)$ — the record being the one that $(\pi, d(\cdot))$ generates — remains in $\mathcal{V}$ for all time. (This is the standard exists-strategy-for-all-realizations form. The quantifier order is essential: the record is generated by the realization, so conditioning the admissible realizations on the record would be circular.) Admissible initial beliefs are the compact information sets $B_0 \subseteq X$ generated by the observation structure — the observation fibres $O^{-1}(y)$, $y \in O(X)$, in the constructions of Sections 3.2–3.3 and 3.6. The existential (non-robust) counterpart $\text{EViab}_{\mathcal{I}}(\mathcal{V})$ is the non-robust contrast class.

We use ERViab throughout, since the disturbance classes of sustainability problems are adverse by construction. The theorems are stated for this robust notion; they do not transfer a fortiori to the non-robust contrast class $\text{EViab}_{\mathcal{I}}(\mathcal{V})$, in which there exist an observation-based policy and *some* admissible disturbance realization under which every compatible trajectory remains in $\mathcal{V}$ for all time. Since the robust kernel is contained in the non-robust one, an observation-based policy may exist there that no robust policy survives.

The thesis is now precise: **the epistemic kernel is the greatest recursively viable collection of information states in the sense of the estimation-space reduction cited in Section 5; common-action compatibility is a**

necessary one-step condition, but not by itself a complete characterization — and the obstruction calculus certifies, mechanism by mechanism, the information states outside it.

2.4 Notation

We collect the principal symbols used throughout the paper and fix the two structural symbols: the **observation structure** is always $\mathcal{I}$ (calligraphic I) and the **institution** of Section 6.4 is always $\mathcal{J}$ (calligraphic J); no other letter serves either role. The state space is $X \subseteq \mathbb{R}^n$; the control set is U (set-valued $U(x)$ at state x); the disturbance set is D (set-valued $D(x)$). The dynamics are $\dot{x} = f(x, u, d)$. The closed constraint set is $\mathcal{V} \subseteq X$, described by finitely many C^1 constraint functions q_j ($j = 1, \dots, m$). The safe-control correspondence at $x \in \mathcal{V}$ is $\mathcal{R}_{\mathcal{V}}(x)$. The observation structure is $\mathcal{I} = (Y, O)$ with observation map $O : X \rightarrow Y$; the information set (belief) at time t is B_t ; the initial belief is B_0 — admissible initial beliefs are the compact information sets the observation structure generates, the observation fibres $O^{-1}(y)$, $y \in O(X)$, in the constructions. The common admissible action set at belief B is $U^B(B) = \bigcap_{x \in B} U(x)$ — the only controls an observation-based policy may select without risking inadmissibility, under the convention, used from Proposition 5 on, that an action outside $U(x)$ at a compatible state is itself failure; the common safe-action set is $\mathcal{R}_{\mathcal{V}}^B(B) = \bigcap_{x \in B} \mathcal{R}_{\mathcal{V}}(x)$. The tube-safe action set at review length Δ is $\mathcal{A}_{\text{tube}}(B, \Delta) = \{a : \text{Reach}_{[0, \Delta]}(B, a) \subseteq \mathcal{V}\}$, where $\text{Reach}_{[0, \Delta]}(B, a)$ is the *robust* — all-disturbance, all-solution — reachable set of the system from B under the held action a over $[0, \Delta]$. The first informative observation time is T_{obs} : the first time the observation record discriminates between compatible branches — before it, the branches of Theorem 3’s (H3.1) produce identical records. The adverse-selection correspondences of the exit certificates are $D_{\varepsilon}(x, u) = \{d \in D(x) : D^+q(x; f(x, u, d)) \leq -\varepsilon\}$ (Theorem 5) and $D_{\eta}(x) = \{d \in D(x) : \nabla q_j(x) \cdot f(x, a, d) \leq -\eta/2\}$ (Theorem 2’s proof, local). The upper right Dini derivative of q in direction v is $D^+q(x; v) = \limsup_{h \downarrow 0} [q(x + hv) - q(x)]/h$; for q of class C^1 — the class assumed in the exit theorems — $D^+q(x; v) = \nabla q(x) \cdot v$.

The kernels of the hierarchy are $\text{Viab}(\mathcal{V}; U, \pi_{\text{perf}})$ (perfect information), $\text{RViab}(\mathcal{V})$ (robust), and $\text{ERViab}_{\mathcal{I}}(\mathcal{V})$ (the robust epistemic kernel — Definition 1, the object of this paper), with $\text{proj}_{\exists}(\mathfrak{B}) = \bigcup_{B \in \mathfrak{B}} B$ the existential projection of a collection $\mathfrak{B}$ of information states. Consistency check: when the observation O is injective the record determines the state, so the epistemic and robust kernels agree, $\text{ERViab}_{\mathcal{I}}(\mathcal{V}) = \text{RViab}(\mathcal{V})$, the informational certificates of Sections 3.2–3.3 and 3.6 reduce to the no-merge case, and only the dynamic obstruction of Section 3.5 can certify exclusion — the definitions degenerate to classical robust viability exactly when observation is perfect. The non-robust counterpart $\text{EViab}_{\mathcal{I}}(\mathcal{V})$ is the non-robust contrast class (Section 2.3). The institutionally restricted kernel is $\text{IRViab}_{\mathcal{J}}(\mathcal{V})$ (Section 6.4). The policy classes are π_{perf} (state feedback), Π_{output} , Π_{state} , and Π_{CE} — the certainty-equivalence class defined in Remark 5: fixed perfect-information regulation laws applied uncorrected to the observation. The policy-specific safety set of Section 4.1 is $\text{Safe}_{\mathcal{I}}^{\Pi}$; the certainly-safe set of Corollary 1 is $\mathcal{V}_{\text{safe}}$.

A few letters carry local meanings, as declared here: the strip width a of Theorem 5’s $\mathcal{S}_a$ is not the held action $a = \pi(B)$ of Theorem 2’s proof, nor the pathway vector $a \geq 0$ of Section 5(d); λ is the Farkas multiplier of Theorem 2’s checkability certificate and the observer decay rate of Section 5(c); and α is the comparison function of Remark 4 and a multiplier of Section 5(d)’s substitution alternative. The letter K denotes the safe set of Section 4’s certification problems, while in Section 5(c) it denotes the kernel itself, with the buffered set $K_{\mathcal{C}}$ and eroded

kernels $K^{-\mathcal{C}}$ of Section 5(c). The observation-error map of Section 2.3 is $H(\xi, v)$, whereas the harvest vector of the Supplementary Material is the lowercase h ; and the finite-horizon recursion set $\mathcal{W}_N$ of Section 3.1 is calligraphic, distinct from the Lyapunov function $W = S_1 + S_2$ of the Supplementary Material.

Shared symbols across the companion papers. The companion papers (worked systems, computational certification, probabilistic sufficiency, exact belief computation, minimax dual certificates, applied regime viability, and the applied forecast ladder) deliberately share one formal vocabulary; the letters below carry paper-local meanings here that differ there:

Here	In the companions
K : the safe set of Section 4’s certification problems; the kernel of Section 5(c)	window depth and contact set $\mathcal{K}(B)$ (minimax duals); the full-information kernel K_{full}^T (certification); stock carrying capacities (forecast ladder; minimax benchmark)
λ : the Farkas multiplier of Theorem 3’s certificate; the observer decay rate of Section 5(c)	certificate weight vectors (certification; minimax duals)
Γ_h : the recourse certificate of Section 5(d)	the certificate margin $\Gamma_{\mathcal{I}}$ (certification); the alpha-vector witness sets Γ_k^{Π} (sufficiency)
τ : delay and observation times	the worst-case exit time τ^* (worked systems)
σ : the recourse construction’s trajectory parameter	the kernel time variable (certification; minimax duals); the floored exit count σ^* (worked systems); a noise scale (forecast ladder)
μ : a Farkas multiplier	an adversarial probability measure (certification; minimax duals)

The four correspondences of this section are one skeleton. An observation-based regulator selects a single response for an entire information set — a held action, a feedback segment, or a safe/unsafe verdict — and observation-based viability is the existence of a response that is simultaneously admissible, safe, and recursively viable for every compatible state and disturbance history. Each obstruction certificate of Sections 3 and 4 is a checkable sufficient condition under which one of these response correspondences is empty: $U^B(B)$ (admissibility), $\mathcal{R}_{\mathcal{V}}^B(B)$ (instantaneous safety), $\mathcal{A}_{\text{tube}}(B, \Delta)$ (review-interval safety), or the recursive predecessor of Section 3.1. The mechanisms differ in which correspondence fails, and the obstructions are nested (Proposition 3). The organizing claim of the calculus is the following, stated once and used implicitly by every certificate of Sections 3 and 4.

Proposition 1 (selector principle). If $B \in \text{ERViab}_{\mathcal{I}}(\mathcal{V})$ and π is a policy witnessing viability on B , then at every review time the action $a = \pi(B)$ selected by π is a response that is (i) *admissible*, $a \in U^B(B)$; (ii) *safe*, keeping every compatible trajectory in $\mathcal{V}$ until the next observation — $\text{Reach}_{[0, \Delta]}(B, a) \subseteq \mathcal{V}$ at review length Δ ; and (iii) *recursively viable*, in that every post-observation belief reached at the next review is again in $\text{ERViab}_{\mathcal{I}}(\mathcal{V})$ under the continuation of π . A belief is therefore viable only if each of the response correspondences of this section — $U^B(B)$, $\mathcal{R}_{\mathcal{V}}^B(B)$, $\mathcal{A}_{\text{tube}}(B, \Delta)$, and the recursive predecessor of Section 3.1 — is nonempty along every reachable branch; hence a certificate that any one of them is empty at a reachable belief certifies nonviability. In finite systems the converse is exact (Theorem 1); in continuous time the converse is the backward-recursion completion noted in Section 6.5.

3. The Obstruction Calculus

3.1 Finite-horizon completeness (finite systems)

In the finite setting the calculus is complete: backward recursion over beliefs — the set-membership analogue of the belief-state recursion of partially observable Markov processes (Smallwood and Sondik, 1973) — is sound and complete on any finite horizon, and nonviability is certified by a finite obstruction tree. The certificates of Sections 3.2–3.3 and 3.5–3.6 are sufficient conditions, not a complete characterization of the epistemic kernel.

Fix finite state, action, disturbance, and observation spaces X, A, D, Y , a transition map $x^+ = F(x, a, d)$, an observation map $y = O(x, a, d, x^+)$, a safe set $\mathcal{V} \subseteq X$, and admissible-action sets $U(x) \subseteq A$. For a belief $B \subseteq X$, an action a , and an observation y , the post-belief is

$$\text{Post}(B, a, y) = \{x^+ \in X : \exists x \in B, d \in D(x), \\ x^+ = F(x, a, d), y = O(x, a, d, x^+)\},$$

and y is possible under (B, a) if $\text{Post}(B, a, y) \neq \emptyset$. With $U^B(B) = \bigcap_{x \in B} U(x)$, define the one-step predecessor of a collection $\mathcal{C}$ of beliefs by

$$\text{Pre}(\mathcal{C}) = \{B \subseteq \mathcal{V} : \exists a \in U^B(B) \text{ such that} \\ \text{Post}(B, a, y) \in \mathcal{C} \text{ for every possible } y\},$$

and the finite-horizon kernels by $\mathcal{W}_0 = \{B : B \subseteq \mathcal{V}\}$, $\mathcal{W}_{k+1} = \text{Pre}(\mathcal{W}_k)$. The N -step common safe-action set at a belief B is $\mathcal{A}_N(B) = \{a \in U^B(B) : \text{Post}(B, a, y) \in \mathcal{W}_{N-1} \text{ for every possible } y\}$, so that $B \in \mathcal{W}_N \iff \mathcal{A}_N(B) \neq \emptyset$. The recursion is exact but of worst-case complexity exponential in $|X|$ — the belief space contains up to $2^{|X|}$ beliefs — so finite-checkability is per belief, not a polynomial-time global procedure.

Theorem 1 (finite-horizon soundness and completeness). An initial belief B_0 admits an observation-based policy that keeps every compatible trajectory in $\mathcal{V}$ for N steps if and only if $B_0 \in \mathcal{W}_N$.

Proof sketch. Both directions are inductions on the horizon. Necessity: a policy safe for N steps picks $a \in U^B(B_0)$, and every possible successor belief is safe for $N - 1$ steps, so $B_0 \in \text{Pre}(\mathcal{W}_{N-1})$. Sufficiency: choose the witness action at each step, and every reachable state remains in $\mathcal{V}$ for N steps. If $B_0 \notin \mathcal{W}_N$, the failure is certified by a finite obstruction tree alternating regulator and nature turns — the complete construction, with the one-step abstraction of Example 1 as its smallest tree, is given in the Supplementary Material (S1; the tree is drawn in Figure S2).

3.2 The instantaneous common-action obstruction

The instantaneous obstruction is the static core of the calculus: it certifies nonviability at a single information state, without a timing or horizon argument.

Theorem 2 (common-action obstruction). Let $B \subseteq \mathcal{V}$ be an information set of the specified observation structure (the safe-control correspondence $\mathcal{R}_\mathcal{V}$ is defined on $\mathcal{V}$, so the belief is taken inside it). Suppose the following hypotheses hold.

(H2.1) The selected action is held until the next observation (sampled review: no informative observation arrives before the next action must be selected, and the policy holds the selected action over the review interval — the institutional reading; without a positive dwell time the instantaneous statement below must be replaced by the tube-safety statement below).

(H2.2) Either the **common admissible action set** is empty,

$$U^B(B) = \bigcap_{x \in B} U(x) = \emptyset, \quad (\text{admissibility obstruction})$$

or the **common safe-action set** is empty while common admissibility holds,

$$\mathcal{R}_\mathcal{V}^B(B) := \bigcap_{x \in B} \mathcal{R}_\mathcal{V}(x) = \emptyset, \quad (1)$$

with $U^B(B) \neq \emptyset$ (safety obstruction)

(H2.3) **Closed-loop existence of the local adverse selection (safety case).** The disturbance correspondence is lower semicontinuous, or locally constant near the boundary point at which the obstruction fires, so that the local adverse selection of the proof is realized along the compatible trajectory; alternatively, the certificate is read against the convexified (relaxed) inclusion, which preserves the signed drift (for $C^1 q$ the directional derivative is affine in the velocity).

Then, under hypotheses (H2.1)–(H2.3), $B \notin \text{ERViab}_\mathcal{T}(\mathcal{V})$: every compatible state may be individually robustly viable, and the belief is nevertheless nonviable, because the state-specific safe actions are incompatible.

Proof sketch. If the selected action is inadmissible at some compatible state, the policy fails there outright. Otherwise $a \in U^B(B)$, and the emptiness of the common safe-action set is caused by the boundary: some $\bar{x} \in B \cap \partial\mathcal{V}$ has $a \notin \mathcal{R}_\mathcal{V}(\bar{x})$, giving an active constraint j and a disturbance $\bar{d}$ with $\nabla q_j(\bar{x}) \cdot f(\bar{x}, a, \bar{d}) < 0$. A measurable local adverse selection (closed graph of D , continuity of the drift; realized by (H2.3)) then drives q_j strictly downward from $q_j(\bar{x}) = 0$ within the first review period, before any observation can arrive.

For polyhedral data the emptiness certificate is a Farkas pair — a finite, independently checkable certificate of epistemic infeasibility — and for infinite beliefs the Helly-type witness of Proposition 4 supplies a finite-witness reduction under the convexity hypotheses of Section 3.4; the complete discussion is given in the Supplementary Material (S1).

Remark 1 (numerical robustness of the polyhedral certificate). Linearity makes the Farkas certificate robust in two practical senses. First, bounded data perturbation degrades the margin gracefully: if $\mu \geq 0$ certifies $\mu^\top A = 0$ and $\mu^\top b \leq -\mu_0 < 0$, then for perturbed right-hand sides $b + \Delta b$ with $\|\Delta b\|_\infty \leq \epsilon$ the same multipliers give $\mu^\top (b + \Delta b) \leq -\mu_0 + \|\mu\|_1 \epsilon$, still negative while $\epsilon < \mu_0 / \|\mu\|_1$; the margin μ_0 is exactly the certificate’s numerical robustness, and normalized multipliers ($\|\mu\|_1 = 1$, as in Section 3.4’s worked certificate) maximize the tolerable perturbation. Second, verified approximate data suffice: if the multipliers are computed with residual $r := \hat{\mu}^\top \hat{A}$ and objective $c := \hat{\mu}^\top \hat{b}$, then feasibility of some $u \in U$ would imply $-h_U(-r) \leq r^\top u \leq c$, where $h_U(g) := \sup_{v \in U} g^\top v$ is the support function of the action set; hence the test $c + h_U(-r) < 0$ — with all integration, rounding, and enclosure errors charged into c and r before it is applied — certifies infeasibility without exact stationarity. Both extensions share the orientation rule for nonviability certificates: uncertain row data must be relaxed toward a larger feasible controller set, never toward a smaller one; tightening $\hat{b}$ on the strength of an unverified worst case can manufacture a false obstruction.

The tube-safety form. The instantaneous condition (1) is the $\Delta \downarrow 0$ boundary approximation of the exact one-step obstruction: with the action held over a review interval of length Δ , the exact condition is the emptiness of the tube-safe action set

$$\mathcal{A}_{\text{tube}}(B, \Delta) = \{a : \text{Reach}_{[0, \Delta]}(B, a) \subseteq \mathcal{V}\},$$

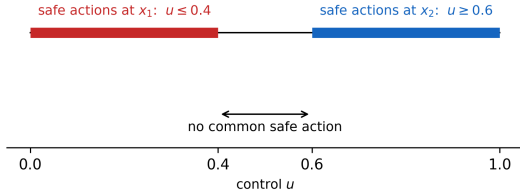


Figure 1: Incompatible safe controls. The states x_1 and x_2 share an observation, but their safe-action sets $[0, 0.4]$ and $[0.6, 1]$ are disjoint, so no common safe action exists (Theorem 2).

with $\text{Reach}_{[0, \Delta]}(B, a)$ the robust (all-disturbance) reachable set (the reading specified in Section 2.4), and $\mathcal{A}_{\text{tube}}(B, \Delta) = \emptyset$ certifies $B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$ for every review interval of length Δ and every target collection. The theorems above and below use the instantaneous and timing readings; the tube form is the unifying one-step object.

Proposition 2 (uniform margin gives a tube obstruction at every review length). Let $B \subseteq \mathcal{V}$ be a compact information set with $U^B(B) \neq \emptyset$, f and ∇q_j continuous, and suppose there is $\eta > 0$ such that for every $a \in U^B(B)$ there exist a boundary state $x_a \in B \cap \partial \mathcal{V}$, an active constraint j with $q_j(x_a) = 0$, and a disturbance $d_a \in D(x_a)$ satisfying

$$\nabla q_j(x_a) \cdot f(x_a, a, d_a) \leq -\eta.$$

Under the closed-loop existence convention of Theorem 2, the tube-safe action set is then empty at every review length,

$$\mathcal{A}_{\text{tube}}(B, \Delta) = \emptyset \quad \forall \Delta > 0,$$

and hence $B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$ under sampled review of any length.

Proof sketch. For fixed $a \in U^B(B)$, continuity of the drift and the closed graph of D extend the margin to $-\eta/2$ on a neighbourhood of (x_a, a, d_a) , and a measurable local selection realizes it; the trajectory from x_a under the held action then satisfies $q_j(x(t)) \leq -(\eta/2)t < 0$ for small t , so $a \notin \mathcal{A}_{\text{tube}}(B, \Delta)$ for every $\Delta > 0$. As a was arbitrary, the tube-safe action set is empty.

Proposition 3 (obstruction ladder). Under the standing regularity assumptions and the closed-loop existence convention of Theorem 2, the common response sets of Section 2.4 are nested:

$$\mathcal{A}_{\text{tube}}(B, \Delta) \subseteq \mathcal{R}_{\mathcal{V}}^B(B) \subseteq U^B(B) \quad \forall \Delta > 0,$$

and emptiness descends the ladder:

$$\begin{aligned} U^B(B) = \emptyset &\implies \mathcal{R}_{\mathcal{V}}^B(B) = \emptyset \implies \mathcal{A}_{\text{tube}}(B, \Delta) = \emptyset \\ &\implies B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V}). \end{aligned}$$

Proof sketch. The inclusion $\mathcal{R}_{\mathcal{V}}(x) \subseteq U(x)$ gives the second nesting. For the first, an action outside $\mathcal{R}_{\mathcal{V}}^B(B)$ violates $\nabla q_j(x) \cdot f(x, a, d) \geq 0$ at some active boundary state, and the corresponding trajectory leaves $\mathcal{V}$ in arbitrarily small time, so the action is not tube-safe.

The obstruction ladder’s four rungs — admissibility, common safe-action, tube, and recursive — are itemized and shown strict in general in the Supplementary Material (S1, Figure S1); the dynamic exit certificate of Theorem 5 sits below the ladder, emptying a response set at some compatible state even under full information.

Example 1 (hidden-mode conflict). Let an unobserved parameter satisfy $\theta \in \{-1, +1\}$, with $\dot{z} = \theta u$, $u \in \{-1, +1\}$, and constraint $z \geq 0$. At $z = 0$: if $\theta = +1$, only $u = +1$ is safe; if $\theta = -1$, only $u = -1$ is safe. Both compatible states are individually robustly viable under full information (each admits its safe control), but the information set $B = \{(0, +1), (0, -1)\}$ satisfies $\mathcal{R}_{\mathcal{V}}^B(B) = \{+1\} \cap \{-1\} = \emptyset$, so $B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$. This is a purely informational failure: no stochasticity and no estimation-quality argument is involved. The example is the two-action skeleton of every “the stock is either recovering or collapsing, and the safe policy differs between the two” situation.

3.3 The delayed-information obstruction

Theorem 3 (delayed-information obstruction). Let B_0 be an initial information set. Suppose the following hypotheses hold.

(H3.1) No informative observation arrives before time $T_{\text{obs}} > 0$ — formally, there are branches that are **observation-equivalent** on $[0, T_{\text{obs}})$: they produce identical observation records throughout that interval, so the policy cannot distinguish them before T_{obs} ; between observations the information set evolves by the set-membership semantics of Section 2.3.

(H3.2) There exist a constraint function q of class C^1 (the class of Theorem 5’s proof, Section 3.5) with $\mathcal{V} \subseteq \{q \geq 0\}$ and a constant $\varepsilon > 0$ such that, **for every implementable blind-window control** — every measurable $u(\cdot)$ on $[0, T_{\text{obs}})$ that is admissible at every compatible state throughout the blind window, $u(t) \in U^B(B_t)$ with B_t the set-membership belief propagated under $u(\cdot)$ — which is exactly the class of controls that an observation-based policy can realize before the first informative observation, where its actions are a fixed function of time — there are a compatible initial state $x^* \in B_0$ attaining $q(x^*) = \inf_{x \in B_0} q(x)$ (with B_0 compact and q lower semicontinuous; otherwise replace the infimum by $\inf q + \delta$ for an arbitrary δ -minimizer, let $\delta \downarrow 0$, and read the timing bound with $\inf q + \delta$) and an admissible disturbance realization such that, along the realized trajectory from x^* under that open-loop control, the record remains compatible and

$$D^+q(x(t); f(x(t), u(t), d(t))) \leq -\varepsilon \quad \text{while } q(x(t)) > 0, \quad (2)$$

(H3.3) The timing bound holds:

$$T_{\text{obs}} > \frac{\inf_{x \in B_0} q(x)}{\varepsilon}. \quad (3)$$

Then, under hypotheses (H3.1)–(H3.3), $B_0 \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$: information may be accurate but arrive too late.

Proof sketch. By (H3.1) the policy’s actions on $[0, T_{\text{obs}})$ form one fixed open-loop control; if it is ever inadmissible at a compatible state the policy fails outright, otherwise (H3.2), applied to this implementable control, supplies a compatible state x^* with $q(x^*) = \inf_{B_0} q$ and an admissible disturbance realizing the drift (2); integrating (2) gives $q(x(t)) \leq \inf_{B_0} q - \varepsilon t$, so the violation $q < 0$ occurs by time $\inf_{B_0} q / \varepsilon$, which (3) places strictly before T_{obs} — before any informative observation, so the policy cannot react in time.

Hypothesis (2) is a condition on the data of the problem, not on the policies, and condition (3) is the timing bound: inverted, it is the design requirement that the first informative observation precede the enforced exit time. The complete discussion of both hypotheses, and the formal-zero-margin comparison with Theorem 2, are given in the Supplementary Material (S1).

Remark 2 (threshold form of the timing obstruction). The delayed-information obstruction has a sharp threshold. Let $\sigma^*(B_0)$ be the guaranteed blind-window survival time: the

largest duration such that some pre-observation policy keeps every observation-equivalent branch in $\mathcal{V}$ until then,

$$\sigma^*(B_0) = \sup_{u(\cdot)} \inf_{x_0 \in B_0, d(\cdot)} \tau(x_0, u, d),$$

with the infimum over branches that remain observation-equivalent on $[0, T_{\text{obs}}]$ and τ the first violation time under the convention of Theorem 3. Then

$$\sigma^*(B_0) < T_{\text{obs}} \implies B_0 \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V}),$$

because before T_{obs} a policy cannot condition its action on which branch realizes. Condition (3) certifies this threshold: under (H3.1)–(H3.2) every branch exits by time $\inf_{x \in B_0} q(x)/\varepsilon$, hence $\sigma^*(B_0) \leq \inf_{x \in B_0} q(x)/\varepsilon$, and (3) states that this upper bound lies below T_{obs} ; the threshold itself may be strictly smaller, in which case nonviability holds even when (3) fails. The threshold is the blind-window response correspondence of the timing layer: its emptiness is the timing certificate, the delayed-information analogue of the tube and common-action correspondences.

Checkability of the timing certificate. The drift hypothesis (3) is finitely checkable whenever the blind-window control class is finite: if B_0 is finite and the admissible blind-window controls form a finite family (hold-until- T_{obs} controls with u in a finite set, or piecewise-constant controls on a finite partition), the certificate reduces to the finite check that every candidate control enforces exit before T_{obs} , each worst-case exit time computed by branchwise integration or by the recursion of Section 3.1. The hidden-regime instance of Section 9 is this check in closed form: the worst branch declines at unit rate, giving $\sigma^*(B_0) = z_0 - 1$, and the certificate fires exactly when $T_{\text{obs}} > z_0 - 1$. For continuous control classes the check remains a template, part of Open Problem 1; over polytope-declared classes, however, the check is exact and finite (Theorem 4), and the class declaration itself is load-bearing (Example 2).

Theorem 4 (linear-programming instantiation of the timing certificate). Let the blind window have K discrete steps, finitely many branches $j \in J$ with robust affine dynamics $x^+ = A_j x + B_j u + E_j w$, $w \in W_j$ a polytope, known initial branch states $x_{j,0}$, polyhedral safe set $\mathcal{V} = \{x : Fx \leq g\}$, and declared blind-window class $\Pi_B = \{(u_0, \dots, u_{K-1}) : u_k \in U\}$ for a polytope U . Then: (i) $\sigma^*(B_0; \Pi_B)$ is computed exactly by at most $\lceil \log_2 K \rceil + 1$ linear programs of size polynomial in K , $|J|$, and the description lengths of U , W_j , $\mathcal{V}$ — unrolling the dynamics makes each robust floor row one linear inequality in the control sequence, the disturbance entering only through precomputed support constants — and with rational data every step is exact. (ii) Infeasibility at horizon K is certified by Farkas multipliers: nonnegative weights on the rows whose aggregate control projection vanishes and whose aggregate constant is negative — the timing certificate in adjoint-row form. (iii) The program’s feasible set is the declared class itself, so the computation is sound and complete over Π_B : no relaxation gap arises. (iv) The declaration is load-bearing: the same computation over a polytope $U \supseteq U_{\text{fin}}$ holding a finite implementable class computes the relaxation, which remains sound but can be arbitrarily incomplete — on the hidden-regime data with $U_{\text{fin}} = \{-1, +1\}$ one has $\sigma^* = z_0 - 1$, while $U = [-1, 1]$ admits $u \equiv 0$, which holds both branches on the floor forever, so $\sigma_{\text{rel}}^* = \infty$ and the relaxed certificate never fires.

Proof sketch. Unrolling the affine dynamics makes each branch state $x_{j,k}$ affine in $(u_0, \dots, u_{k-1})$, so every robust floor row is one linear inequality once $\max_{w \in W_j} FE_j A^{\ell} w$ is precomputed; feasibility of the horizon- K' system is monotone in K' (prefix truncation), giving (i) by bisection, and Farkas’ lemma gives (ii). Claim (iii) is definitional: the program’s variables

are the declared controls themselves. Claim (iv) is the computation displayed in Example 2; the complete proof, including the duality computation, is in Supplementary S1.

Example 2 (the relaxation gap is unbounded). Two branches share $z^+ = \frac{9}{10}z \pm u$ from a common z_0 , floor $\mathcal{V} = \{z \geq 1\}$, declared class $U = [-1, 1]$ per step. The branch sum $z_k^+ + z_k^- = 2(\frac{9}{10})^k z_0$ is invariant under the control, so both floors force $z_0 \geq (\frac{10}{9})^k$ at every step k : the window- K program is feasible exactly when $z_0 \geq (\frac{10}{9})^K$. The level- k Farkas certificate is the pair of floor rows summed — their control coefficients cancel pairwise — leaving $2(\frac{9}{10})^k z_0 \geq 2$; and $u \equiv 0$ attains feasibility on the boundary $z_0 = (\frac{10}{9})^K$, keeping the branches balanced at $(\frac{10}{9})^{K-k} \geq 1$. The program is thus a complete characterization of the declared class. If instead the declared class is the finite set $\{-1, +1\}$, the guaranteed survival time is $\sigma^* = z_0 - 1$: strictly smaller. The relaxation enlarges the class and, on the hidden-regime data, extinguishes the certificate altogether.

Remark 3 (adjoint-row correspondence). The Farkas multipliers of (ii) are the finite analogue of the adjoint safety rows of the continuous-to-finite bridge: a weight $\lambda_{j,k,i}$ on the i -th floor row of branch j at step k plays the role of an adjoint variable on a continuous-time safety row, the vanishing aggregate control projection is the pooled adjoint condition, and the negative aggregate constant is the certificate margin. The control coefficients of Example 2 cancel pairwise within each level, which is the discrete image of the pooled adjoint rows’ cancellation. The bridge to continuous time — error-controlled sampling of these programs — is developed in a companion work.

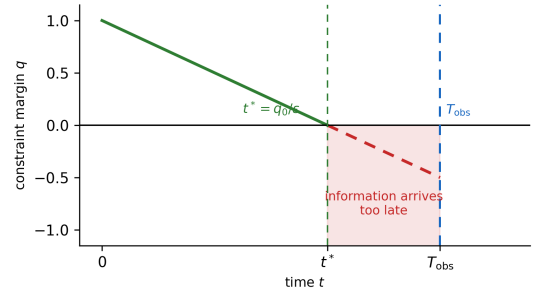


Figure 2: Delayed information. The constraint margin q reaches zero at $t^* = q_0/\varepsilon$, before the first informative observation at T_{obs} : the information is accurate but arrives too late (Theorem 3).

3.4 The sparse finite witness

Proposition 4 (sparse common-action witness). Let $B \subseteq \mathcal{V}$ be an information set, $U(x) \subseteq U$ with $U \subseteq \mathbb{R}^m$ compact and convex and $U(x)$ convex and closed, let the dynamics be affine in the control, $f(x, u, d) = f_0(x, d) + f_u(x, d)u$, with $D(x)$ compact, and let $\mathcal{V}$ have C^1 constraint functions q_j . Then each safe-control set $\mathcal{R}_{\mathcal{V}}(x)$ is a compact convex subset of U , and the safety-case common-action obstruction $\bigcap_{x \in B} \mathcal{R}_{\mathcal{V}}(x) = \emptyset$ holds if and only if it is witnessed by at most $m+1$ compatible states: there exist $x_1, \dots, x_{m+1} \in B$ with $\bigcap_{i=1}^{m+1} \mathcal{R}_{\mathcal{V}}(x_i) = \emptyset$. The admissibility analogue (with $\mathcal{R}_{\mathcal{V}}(x)$ replaced by $U(x)$) holds under the same convexity assumptions.

Proof. For fixed (x, j, d) the inequality $\nabla q_j(x) \cdot f(x, u, d) \geq 0$ is affine in u , so $\mathcal{R}_{\mathcal{V}}(x)$, an intersection of closed halfspaces (over active j and $d \in D(x)$) with the closed convex $U(x)$, is closed and convex; contained in the compact U , it is compact. If the intersection over all $x \in B$ is empty, then by compactness some finite subfamily of $\{\mathcal{R}_{\mathcal{V}}(x)\}_{x \in B}$ already has empty

intersection, and by the finite Helly theorem at most $m + 1$ members suffice; the converse direction is immediate. $\square$

Scope. Both conclusions rest on convexity: the finite Helly theorem is what bounds the witness by $m + 1$, and beyond convex safe-control sets neither the bound nor the finite checkability of the certificate survives in general.

A worked certificate (two floors, one scalar action).

Two floors demand incompatible extraction rates on a single control $u \in [0, 1]$. At the compatible state x_1 (floor 1 active) the safe controls are $u \leq 0.4$; at x_2 (floor 2 active) they are $u \geq 0.6$. The observation merges x_1 and x_2 , so a common safe action must satisfy both. Writing the constraints as $Au \leq b$ with $A = (1, -1)^\top$ and $b = (0.4, -0.6)^\top$, the stacked system $u \leq 0.4$, $-u \leq -0.6$ is infeasible; the Farkas certificate is $\lambda = (1/2, 1/2)$ (normalized, $\|\lambda\|_1 = 1$), for which $\lambda^\top A = 0$ and $\lambda^\top b = -0.1 < 0$. Under the normalization $\|\lambda\|_1 = 1$, the magnitude $|\lambda^\top b| = 0.1$ is the infeasibility margin of the stacked system, shrinking to zero exactly as the two safe-action sets come to touch. The multiplier exhibits the contradiction as a convex combination of the two floor constraints: no common extraction rate respects both floors, and any observation that separates x_1 from x_2 restores one-step viability (Figure 1). With $m = 1$, the two incompatible states are exactly the $m + 1$ -state witness of Proposition 4.

Scope of the convexity hypotheses. Convexity is essential to the witness: without it the bound fails even in a single control dimension. Take $U = \mathbb{R}$ ($m = 1$) and three nonconvex safe-action sets $\mathcal{R}_1 = \{0, 1\}$, $\mathcal{R}_2 = \{0, 2\}$, $\mathcal{R}_3 = \{1, 2\}$. Each pair intersects ($\mathcal{R}_1 \cap \mathcal{R}_2 = \{0\}$, $\mathcal{R}_1 \cap \mathcal{R}_3 = \{1\}$, $\mathcal{R}_2 \cap \mathcal{R}_3 = \{2\}$), yet the three-way intersection is empty, so no $(m + 1) = 2$ -state witness exists. The convexity hypotheses of Proposition 4 exclude exactly this failure mode.

Scope of the count: instantaneous actions, not window-policies. The $m + 1$ bound counts independent instantaneous decisions: it applies to a common action $u \in U \subseteq \mathbb{R}^m$, and it fails for common blind control *functions*. Partition a blind window into Q blocks $I_1, \dots, I_Q$ and let $q + 1$ branches be indistinguishable throughout, with scalar input ($m = 1$): branch $j \leq q$ evolves as $\dot{x}_j = \mathbf{1}_{I_j}(t)(u - 1)$ from $x_j(0) = 0$ under the floor $x_j \geq 0$, which forces $u \equiv 1$ on every block (each $w_j = \int_{I_j} u dt = 1$), while branch $q + 1$ evolves as $\dot{x}_{q+1} = -u$ from $x_{q+1}(0) = q - \varepsilon$ under the same floor, forcing $\sum_j w_j \leq q - \varepsilon$. Summing the forced withdrawals gives $0 \leq -\varepsilon$: the full system is infeasible, while removing any single branch restores feasibility ($u \equiv 1$, or $u = 0$ on the dropped branch's block and 1 elsewhere), so every obstruction involves all $q + 1$ branches — far more than $m + 1 = 2$. The adjoint kernels here are the block indicators, of rank q : for window-policies the governing count is the information–time rank, and sparsity claims for review-interval and delayed-sensing certificates must be stated in the block dimension mQ , not the instantaneous dimension m (cf. the block structure of the delayed-information check in Section 8).

3.5 The finite-time exit certificate

The base obstruction operates before any observation-theoretic argument: it certifies membership in the full-information “shadow” — the complement of the viability kernel (Aubin, 2001; Aubin and Catté, 2002). It is a drift condition under which *no* control — of any information structure — can be viable. Put in management terms, when the dynamics themselves force exit before any control can react, no monitoring design can rescue viability, and the certification problem is closed without observation-theoretic reasoning.

Theorem 5 (finite-time exit certificate). Let $q : X \rightarrow \mathbb{R}$ be a constraint function of class C^1 — the class assumed throughout the exit theorems — with $\mathcal{V} \subseteq \{q \geq 0\}$, and suppose the

following hypotheses hold.

(H5.1) There exist constants $a > 0$ and $\varepsilon > 0$ such that on the strip $\mathcal{S}_a = \{x : 0 \leq q(x) \leq a\}$,

$$\sup_{u \in U(x)} \inf_{d \in D(x)} D^+ q(x; f(x, u, d)) \leq -\varepsilon \quad \forall x \in \mathcal{S}_a, \quad (4)$$

where $D^+ q(x; v) = \limsup_{h \downarrow 0} [q(x + hv) - q(x)]/h$ is the upper right Dini derivative of q in direction v (for C^1 q , $D^+ q(x; v) = \nabla q(x) \cdot v$ — the form the proof uses).

(H5.2) **Closed-loop existence of the adverse realization.** The disturbance correspondence is lower semicontinuous, or constant, on the strip, so that the state-dependent adverse selection constructed in the proof is realized by an admissible control–disturbance pair for which the Carathéodory existence theorem applies. (Alternatively, without (H5.2), the certificate is read against the convexified (relaxed) inclusion $\dot{x} \in \text{co } f(x, u(t), D_\varepsilon(x, u(t)))$, which admits solutions under Marchaud-type conditions and preserves the drift inequality because, for C^1 q , the directional derivative $\nabla q(x) \cdot v$ is affine in the velocity; closure and relaxation are the standard tools for this step.)

Then, under hypotheses (H5.1)–(H5.2), for every admissible control and every initial state $x_0 \in \mathcal{S}_a$ there exists an admissible disturbance realization under which the trajectory attains $q(x(t)) < 0$ — hence leaves $\{q \geq 0\} \supseteq \mathcal{V}$ — by every time strictly larger than $q(x_0)/\varepsilon$: $\inf\{t : q(x(t)) < 0\} \leq q(x_0)/\varepsilon$, in particular within time at most a/ε (since $q(x_0) \leq a$). (Convention, used in both exit theorems: *violation* is the strict inequality $q < 0$; at $q = 0$ the state may still lie on $\partial \mathcal{V} \subseteq \mathcal{V}$.)

Proof sketch. Fix an admissible control. By (4), the adverse-selection correspondence $D_\varepsilon(x, u)$ has nonempty values on the strip; its closed graph and compact values (Section 2.1) yield a measurable selection $d(\cdot)$, realized in closed loop by (H5.2). The fundamental theorem of calculus integrates (4) along the resulting trajectory to $q(x(t)) \leq q(x_0) - \varepsilon t$, so q attains the strict violation $q < 0$ by time $q(x_0)/\varepsilon \leq a/\varepsilon$.

Remark 4 (comparison-function form). Hypothesis (H5.1) uses a constant drift margin ε . The certificate extends to state-dependent decline: if there is a continuous $\alpha : (0, a] \rightarrow (0, \infty)$ with

$$\sup_{u \in U(x)} \inf_{d \in D(x)} D^+ q(x; f(x, u, d)) \leq -\alpha(q(x)) \quad \forall x \in \mathcal{S}_a,$$

then the exit-time bound sharpens to

$$t_{\text{exit}} \leq \int_0^{q(x_0)} \frac{ds}{\alpha(s)},$$

whenever the integral is finite. The proof replaces the linear comparison $q(x(t)) \leq q(x_0) - \varepsilon t$ with the solution r of $\dot{r} = -\alpha(r)$, $r(0) = q(x_0)$, by the standard comparison argument. If $\alpha(0) = 0$ and the integral diverges, the conclusion weakens to asymptotic approach rather than finite-time exit; the constant-margin case $\alpha \equiv \varepsilon$ is the sharp finite-time instance.

Two readings of (4) matter. First, it is an **Isaacs-type condition**: the disturbance chooses after the control, so the certificate asserts that the disturbance can *enforce* exit against any policy. Second, (4) is an outward-drift certificate *dual in spirit*, but *not logically complementary*, to an inward barrier condition. The pointwise robust barrier reading is $\sup_u \inf_d D^+ q \geq 0$ — some control keeps the drift inward against every disturbance — and the two conditions leave an unresolved intermediate region in which neither holds (Section 6.2). Theorem 5 is the “unsafety” side of that duality, and it is unconditional: it needs no observation argument because it defeats all controls, including clairvoyant ones.

3.6 Epistemic emptiness by admissibility: a minimal construction

The admissibility obstruction applies to systems in which every state is individually viable under full information and the kernel empties only under the observation structure. The minimal construction of this section empties the kernel through *admissibility* — a state-dependent control set whose common action is forced and unsafe at the boundary; the common-action obstruction of Section 3.2 and Example 1’s hidden-parameter conflict are the informational cases proper, merging states with incompatible safe controls. In ecological terms, each stock is manageable on its own; the failure arises because the indicator cannot tell the manager which stock is present.

Proposition 5 (epistemic emptiness by admissibility — a minimal construction). There exists a system with $\text{Viab}(\mathcal{V}; U, \pi_{\text{perf}}) = \mathcal{V} \neq \emptyset$ such that, for a non-injective observation map O , every admissible initial information state — the observation fibres $B_0 = O^{-1}(O(S_0))$, $S_0 \in \mathcal{V}$ — lies outside $\text{ERViab}_{\mathcal{I}}(\mathcal{V})$: the epistemic kernel over the fibre-induced initial beliefs is empty. The mechanism is *admissibility* rather than a hidden mode: the state-dependent control set of the construction leaves a single common action at the merged belief — one that is outward-drifting at the boundary state — so the emptiness is the common-action obstruction (Theorem 2’s safety case) at minimal size, produced by the control-set primitive rather than by information; the hidden-mode instance is Example 1.

Proof sketch (construction). Take $X = [1, 2]$, $\dot{S} = u - r(S)$ with r continuous and injective, $U(S) = \{0, r(S)\}$, $\mathcal{V} = [1, 2]$, and the constant observation $O \equiv 0$. Under full information $u = r(S)$ gives $\dot{S} = 0$, so every state is viable. Under $O \equiv 0$ the common control set over the fibre is $\{0\}$ — injectivity of r rules out a common $r(S)$ — so the policy must hold $u = 0$, and $\dot{S} = -r(S) < 0$ drives every compatible trajectory out of $\mathcal{V}$ in finite time.

The construction is the minimal instance of Theorem 2’s admissibility mechanism — with a constant control set in the same plant the fibre is epistemically viable — so the empty-kernel phenomenon is epistemic rather than dynamic; the complete isolation of the mechanism is given in the Supplementary Material (S1).

3.7 A recourse certificate for the post-observation mode

The certificates of the preceding subsections fire before information discriminates: the common-action obstruction is instantaneous (Section 3.2) and the timing obstruction exits within the blind window (Section 3.3, Remark 2). Section 6.5(ii) isolates the complementary mode: insufficient post-observation recourse. In that mode a common blind-window control keeps every branch safe, yet no observation-adapted control can recover every branch once the distinguishing observation arrives; the discrete instance of Supplementary A.3 exhibits the mode while no certificate fires. This subsection supplies a continuous-time certificate for the mode. Its soundness covers every measurable information-adapted policy. Completeness is not claimed, because the certificate relaxes the post-observation class upward: an oracle grants each stored branch its own controller whether or not the sensor reveals that much, and an oracle failure is only a sufficient condition for real failure.

Proposition 6 (oracle-recourse obstruction). Let the dynamics be affine in the control, $f(x, u, d) = f_0(x, d) + B(x)u$, with compact convex $U \subseteq \mathbb{R}^m$, and let branches $x_1, \dots, x_L \in B_0$ be observation-equivalent on $[0, \tau]$ with horizon $T > \tau$ (disturbances admitted, D compact). Fix a label $\ell = (\theta, j, t)$ —

branch θ , constraint j , time $t \in (\tau, T]$ — and a disturbance realization $d(\cdot)$. Let $\bar{x}_\theta^d(\cdot)$ denote branch θ ’s zero-control realization under $d(\cdot)$, $X_\theta^d(\cdot, \cdot)$ its state-transition matrix, and $B^d(\sigma) := B(\bar{x}_\theta^d(\sigma))$. Define the input kernel

$$k_\ell^d(\sigma) = \mathbf{1}_{\{\sigma \leq t\}} (B^d(\sigma))^\top (X_\theta^d(t, \sigma))^\top \nabla q_j(\bar{x}_\theta^d(t)),$$

let $\beta_\ell^d = q_j(\bar{x}_\theta^d(t))$ denote the zero-control facet margin at the label, and write $\phi_U(g) := \min_{v \in U} g^\top v$. For weights $\lambda_\ell \geq 0$, $\sum_\ell \lambda_\ell = 1$, define the recourse certificate

$$\Gamma_h(\lambda) = \inf_{d(\cdot)} \left\{ \sum_\ell \lambda_\ell \beta_\ell^d + \int_0^\tau h_U \left(\sum_\ell \lambda_\ell k_\ell^d(\sigma) \right) d\sigma + \sum_\ell \int_\tau^T h_U(\lambda_\ell k_\ell^d(\sigma)) d\sigma \right\},$$

with $h_U(g) := \max_{v \in U} g^\top v$ the support function of U ; the infimum runs over admissible disturbance realizations (for affine dynamics each bracket is computable in closed form by variation of constants). If $\Gamma_h(\lambda) < 0$, then $B_0 \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$: some stored facet is violated along some branch, for every policy in the stated class and every branch selection.

Proof sketch. Fix a disturbance realization attaining the infimum and any admissible policy. Variation of constants expands the weighted facet sum at the stored labels into the β -term, the window pairing (one common measurable signal on $[0, \tau]$, by the blind-window reduction of Theorem 3), and the branch pairings (separate post-reveal signals); each realized pairing is bounded above by the corresponding support value, so the weighted facet sum is at most $\Gamma_h(\lambda)$. A strictly negative weighted sum forces some facet value strictly negative — the facet is violated along its branch — and the window signal was arbitrary, so every policy in the class fails. The complete proof, the comparison with the attainment-special ϕ_U -variant, and the position of the pooling weights are given in the Supplementary Material (S1). $\square$

Example 3 (three branches, delayed recourse). A planar braking instance in which the obstruction is purely one of post-observation recourse. Branches $j = 1, 2, 3$ run $\dot{p} = v$, $\dot{v} = u$ (no disturbance) from $p_j(0) = \frac{34}{25}n_j$, $v_j(0) = n_j$, where $n_1 = (1, 0)$, $n_2 = (-\frac{3}{5}, \frac{4}{5})$, $n_3 = (-\frac{3}{5}, -\frac{4}{5})$; the actuator set is $U = \text{conv}\{\pm n_1, \pm n_2, \pm n_3\}$ (the hexagon $|u_2| \leq \frac{4}{5}$, $|2u_1 \pm u_2| \leq 2$); safety is $n_i^\top p \leq 2$ for $i = 1, 2, 3$ with $\|v\|_\infty \leq \frac{6}{5}$; the realized branch is revealed at time τ . Every branch is viable under full information — brake along $-n_j$ for one second; peak critical position $\frac{34}{25} + \frac{1}{2} = \frac{93}{50} < 2$ — and $u \equiv 0$ keeps every branch strictly safe through any blind window of length up to $\frac{16}{25}$ ($n_j^\top p_j(\tau) = \frac{34}{25} + \tau$): no instantaneous or blind-window certificate fires. Yet the label $(j, n_j^\top p \leq 2, \tau + 1)$ along branch j has zero-control margin $\beta_j = \frac{16}{25} - \tau - 1$ and kernel $\mathbf{1}_{\{\sigma \leq \tau+1\}}(\tau + 1 - \sigma)n_j$; the weights $\lambda = (\frac{3}{8}, \frac{5}{16}, \frac{5}{16})$ pool the pre-observation kernel to zero, and the certificate evaluates to

$$\Gamma_h(\tau) = \left(\frac{16}{25} - \tau - 1 \right) + \sum_j \lambda_j \int_\tau^{\tau+1} (\tau + 1 - \sigma) d\sigma = \frac{7}{50} - \tau,$$

the pooled pre-observation kernel vanishing under λ and each post-reveal branch’s pairing bounded by its support value ($\sum_j \lambda_j/2 = \frac{1}{2}$). For $\tau > \frac{7}{50}$ the certificate fires and the triple is nonviable — the blind window accumulates braking liability that the post-observation authority cannot discharge on every branch at once — while the hold-then-brake policy ($u = 0$, then $-n_j$ for one second) is safe exactly to $\tau = \frac{7}{50} = 0.14$: the threshold is exact. The obstruction is genuinely three-way: at $\tau = \frac{1}{5}$, every pair of branches is jointly viable under a common blind control (worst critical positions $\frac{2469}{1250} = 1.9752$ and $\frac{2419}{1250} = 1.9352$, both below 2), so six of the seven nonempty priors over these branches are viable at a delay where the triple is

not. The discrete analogue in which no certificate fires is Supplementary A.3; here the certificate fires, $\Gamma_h(\frac{1}{5}) = -\frac{3}{50} < 0$.

The certificate is deliberately asymmetric, like the calculus it joins: it is sound against every measurable policy of the information class, its stored labels are finite and independently checkable (exact arithmetic throughout the instance above), and its incompleteness is explicit — a complete continuous-time characterization of the recourse mode remains open (Section 6.5(ii)).

4. Certification Limits

4.1 Exact certification and the fibre criterion

The obstruction theorems of Section 3 concern *policies*. Sustainability governance also uses *certificates*: verdicts — safe or unsafe — computed from observations alone, without reference to a policy. The question is when such verdicts can be exact. For a manager operating under partial ecological information, the question is whether an indicator reading alone can ever settle whether the system is on the safe side of a specified floor.

Definition 2 (exact certifier). For an admissible domain $Z \subseteq X$ and a safe set $K \subseteq Z$, an **exact certifier** based on the observation map $O : Z \rightarrow Y$ is a function $C : Y \rightarrow \{0, 1\}$ such that

$$C(O(z)) = 1 \iff z \in K \quad \text{for every } z \in Z.$$

Proposition 7 (observation-fibre criterion). An exact certifier based on O exists if and only if membership in K is constant on every observation fibre:

$$O(z_1) = O(z_2) \implies [z_1 \in K \iff z_2 \in K] \quad \forall z_1, z_2 \in Z, \quad (5)$$

equivalently, $K = O^{-1}(O(K))$ on the admissible domain.

Proof sketch. If C exists and $O(z_1) = O(z_2)$, then $\mathbf{1}_K(z_1) = C(O(z_1)) = C(O(z_2)) = \mathbf{1}_K(z_2)$, so membership is fibre-constant. Conversely, (5) makes $C(y) = \mathbf{1}_K(z)$ for any z with $O(z) = y$ well defined and exact.

The fibre criterion concerns static classification — whether membership in a safe set factors through the observation — and applies verbatim to three distinct certification problems, which should not be conflated: constraint certification ($K = \mathcal{V}$), robust viability certification ($K = \text{RViab}(\mathcal{V})$), and policy-specific safety certification ($K = \text{Safe}_T^\Pi$). The same theorem governs all three, but its governance meaning differs. The criterion is the zero-step, label-valued analogue of the common-action test: an observation-based verdict assigns a single label to an entire fibre, exactly as an observation-based action assigns a single action to an entire belief, so both are selector problems. They remain distinct — a fibre may fail exact certification while a policy is still viable, and exact static certification does not imply dynamic viability; hence certification is treated as a layer separate from viability throughout.

Corollary 1 (safety-crossing fibres and the certainly-safe set). If two admissible states share an observation and lie on opposite sides of the *same* component safety constraint — with the safe set of that certification problem read as the single floor $K = \{q_j \geq 0\}$; or, in the general form of the hypothesis, one of the two states lies in K and the other outside K — no exact observation-only certificate exists. (Crossing one floor of an intersection $K = \bigcap_j \{q_j \geq 0\}$ does not by itself put the pair on opposite sides of the intersection — the state satisfying floor j may violate floor i — hence the single-floor scoping.) The largest set of observations that can soundly be labeled safe without further information is the certainly-safe set

$$\mathcal{Y}_{\text{safe}} = \{y \in O(Z) : O^{-1}(y) \subseteq K\}.$$

Proof sketch. A fibre containing one state in K and one outside violates (5), so Proposition 7 denies the certifier. On $\mathcal{Y}_{\text{safe}}$ every compatible state is safe, so the constant verdict “safe” is sound; outside it some compatible state is unsafe, so no sound “safe” verdict exists.

The governance reading: a composite index whose fibres cross a floor boundary cannot support an exact certification of that floor, however carefully it is thresholded; the certainly-safe set is the sound region where the index may certify safety, outside which it must fall silent. The complete discussion, with the fibre figure, is given in the Supplementary Material (S1).

4.2 The certainty-equivalence trap

The obstruction of Proposition 5 relies on a non-injective observation — information lost. The same emptying occurs with a fully *injective* observation for a single policy: the canonical certainty-equivalence controller that applies a perfect-information law to the uncorrected observation.

Remark 5 (the certainty-equivalence trap). Consider $\dot{S} = u - g(S)$ with $u \in [0, \bar{u}]$, g strictly increasing on $[S_{\min}, S^* + b]$, $g(0) = 0$, and $\mathcal{V} = [S_{\min}, S^*]$. Assume the range condition $g(S + b) \leq \bar{u}$ for every $S \in [S_{\min}, S^*]$, so that the controller below is admissible. Under perfect information the zeroing feedback $u(t) = g(S(t))$ gives $\dot{S} = 0$, so every $S_0 \in \mathcal{V}$ is viable and $\text{Viab}(\mathcal{V}; U, \pi_{\text{perf}}) = \mathcal{V} \neq \emptyset$. Now take the injective, biased observation $\hat{S} = S + b$ with $b > 0$, and the canonical certainty-equivalence controller: the perfect-information zeroing law applied directly to the uncorrected observation, $u = g(\hat{S}) = g(S + b)$. Then

$$\dot{S} = g(S + b) - g(S) > 0 \quad \forall S,$$

and since g is strictly increasing on the compact interval $[S_{\min}, S^* + b]$, $\dot{S}$ is bounded below by a positive constant there; hence S strictly increases and exits above S^* in finite time from every $S_0 \in \mathcal{V}$: the kernel of this single controller is empty. The bias-corrected controller $u = g(\hat{S} - b) = g(S)$ uses the observation map’s structure, restores $\dot{S} = 0$, and keeps the kernel nonempty.

The mechanism is a *failure of one policy to use the observation map’s structure*, not a loss of information: for a known invertible observation the output-feedback class is as powerful as the state-feedback class. The phenomenon is the viability-theoretic analogue of Witsenhausen’s counterexample in stochastic control (Witsenhausen, 1968): the information pattern — not the dynamics and not the constraint — defeats the uncorrected controller. The remark is policy-specific; it does not assert that the entire certainty-equivalence class empties the kernel.

5. The Sufficiency Landscape

For contrast, we record the sufficiency results against which the obstruction calculus is defined; the proofs are in the cited sources.

(a) **Veliov’s output-feedback condition.** Veliov (1993) gives a sufficient condition for the existence of an output-feedback regulation map under incomplete and inexact measurement, reducing under perfect measurement to the classical viability condition (Haddad, 1981). Veliov’s theorem certifies existence; the obstruction calculus certifies nonexistence; a problem satisfying neither requires a stronger observer or a finer observation structure.

(b) **The estimation-tube programme.** Cardaliaguet, Quincampoix, and Saint-Pierre (2007) pass from imperfect information in the measurement space to perfect information in an estimation space of sets of compatible states, with equality of value functions and a Dini-derivative characterization.

Under the set-membership semantics the estimation-space solution propagates the belief under a single control signal, so its controls are common controls; the common-action obstruction (Theorem 2) is the certificate that no jointly admissible selection exists at the information state in question (Section 6.3).

(c) Observer-and-buffer transfer. If a full-information feedback exists with a strict inward margin and an observer supplies exponentially convergent estimates, output feedback of the estimate preserves safety on a buffered subset (observer-to-viability transfer with safety buffer; Sontag, 1998), and eroded kernels remain invariant under bounded estimation and implementation errors (erosion absorbs the error). Under margins and convergence, output feedback can work; Sections 3–4 state when it cannot.

(d) The linear substitution alternative. For a finite linear model in which a resource-typed system must meet a demand vector through specified substitution pathways, exactly one of the following holds (Farkas, 1902; Gale, 1960): (i) there exists a non-negative pathway vector $a \geq 0$ satisfying the linear substitution constraints; or (ii) there exist multipliers $\alpha, \beta, \gamma \geq 0$ with

$$\begin{aligned} \alpha^\top R + \beta^\top E - \gamma^\top Q &\geq 0 \quad \text{componentwise,} \\ \gamma^\top s^{\text{req}} &> \alpha^\top x + \beta^\top e. \end{aligned}$$

The second statement is a separation certificate that the pathways cannot meet demand within the typed resource and capacity bounds — the feasibility-side complement of the observation-side obstructions of Sections 3–4.

6. Discussion

6.1 The shape of the calculus

The six mechanisms form a nonexhaustive taxonomy of information-theoretic failure. Table 1 collects the certificates, the response correspondence each one empties, and the design consequence each one licenses.

The exit certificate (Theorem 5) is the dynamic obstruction: it defeats all controls, so it needs no observation argument. The epistemic-emptiness construction (Proposition 5) exhibits the mechanism at minimal size — one scalar state, two admissible controls, one constant observation — in the admissibility form of Section 3.6. The common-action obstruction (Theorem 2) is the static obstruction: it defeats all policies at a single information state. The delayed-information obstruction (Theorem 3) interpolates between them in time. The fibre criterion (Proposition 7) is the certification obstruction: it limits not policies but verdicts. The certainty-equivalence trap (Remark 5) limits a single uncorrected policy. Finite-horizon completeness (Theorem 1) supplies the exact finite-setting counterpart of these sufficient certificates.

A governance failure of observation can be diagnosed by which certificate applies. If the exit certificate holds, no monitoring design helps — the dynamics are adverse under every control. If only the common-action obstruction holds, the remedy is a finer observation structure (an informative measurement distinguishing the incompatible states). If the delayed-information obstruction holds with a slack timing bound, the remedy is a shorter review interval or a faster indicator. If the fibre criterion fails, the remedy is an observation that separates the sides of the floor rather than a better threshold on the aggregate.

6.2 Relation to barrier certificates

Barrier certificates certify safety from a scalar function whose zero level set separates the unsafe region from all trajectories (Prajna and Jadbabaie, 2004; Prajna, Jadbabaie, and Pappas,

2007); the converse, under convex-duality conditions on density functions, is Prajna and Rantzer (2005), and necessary-and-sufficient hybrid characterizations are Maghenem and Sanfelice (2019). Theorem 5 is the unsafety complement of that converse, not its observation-theoretic counterpart: a Dini-drift certificate of unsafety needing no observation structure, constructive in the enforcing disturbance and exit time. The remaining certificates (Proposition 5, Theorem 2, Theorem 3, Proposition 7) act at the level of the information set itself: barrier constructions presuppose a state — or a specified state estimate — for feedback, whereas the common-action and timing obstructions certify the emptiness of the common safe-action set at a given information state, which barrier methods do not address.

6.3 Relation to the estimation-tube programme

The estimation-tube programme shows that imperfect measurement can be absorbed into set-valued dynamics on an estimation space with no loss in value (Cardaliaguet, Quincampoix, and Saint-Pierre, 2007); the present theorems delimit its reach rather than contradict it. The reduction preserves values, and its belief-state controls are common controls; the obstruction calculus is the local certificate layer of that programme — the common-action obstruction certifies that the belief-state predecessor is empty, the timing bound quantifies when it fails for reasons of information arrival, and the fibre criterion states when no observation-based verdict can be exact. Belief-space viability answers *whether*; the obstruction calculus answers *why not, and what to change*. A single worked system carrying the three methods side by side — barrier certificate, estimation-tube reduction, and obstruction certificate — is given in the Supplementary Material (S2).

6.4 Consequences for monitoring and indicator design

Five consequences follow directly from the certificates. Each translates a formal impossibility into a concrete design specification for monitoring systems under partial ecological information.

Timing. Theorem 3 gives the minimal monitoring frequency in closed form: the first informative observation must precede the enforced exit time $\inf q/\varepsilon$. A review interval longer than the worst-case time from the most-constrained compatible state — the minimum- q state, which has the least constraint slack — to constraint violation is an information structure under which the violation is undetectable in time, whatever the review then concludes. The bound is computable from the drift certificate (2), which a robustness analysis computes in any case.

Coarseness. Theorem 2 states that an observation structure merging two compatible states with incompatible safe controls is nonviable *at that information state*, even though both states are viable in isolation. The design consequence is that the observation must separate safety classes, not states: an indicator is exposed to Theorem 2’s obstruction when some observation fibre crosses the safe-control partition of the state space — merging states whose safe controls differ is what the theorem certifies, and keeping every fibre within a single class is what removes that exposure; refinement is never harmful (a finer observation merges nothing the coarser one did not), though Theorem 2 claims no benefit beyond the removal, and other mechanisms may still certify nonviability on a finer partition.

Aggregation. Proposition 7 and Corollary 1 apply verbatim to composite indices, which are observation maps from the system state to a scalar. An index whose fibres cross a floor boundary cannot support exact floor certification; the

Certificate	Emptied response correspondence	Design consequence (Section 6.4)
finite-time exit (Theorem 5)	the response set of some compatible state, even under full information	none — dynamics, not information
admissibility (Proposition 5)	$U^B(B) = \emptyset$	enlarge the command set
common action and tube (Theorem 2, Proposition 2)	$\mathcal{R}_V^B(B) = \emptyset$, $\mathcal{A}_{\text{tube}}(B, \Delta) = \emptyset$	a separating observation; a shorter review interval
delayed information (Theorem 3, Remark 2)	$\sigma^*(B_0) < T_{\text{obs}}$	an earlier informative observation
fibre certification (Proposition 7, Corollary 1)	$K \neq O^{-1}(O(K))$	a refined index; per-floor measurement
certainly equivalence (Remark 5)	kernel of the uncorrected controller $u = g(\hat{S})$ is empty	correct the known bias
finite-horizon recursion (Theorem 1)	$B_0 \notin \mathcal{W}_N$	exact characterization (Section 3.1)

Table 1: The obstruction certificates, the response correspondence each one empties, and the design consequence each one licenses. The first six are sound sufficient conditions for nonviability; the last is the exact finite-horizon characterization.

certainly-safe set is the domain in which an index-based verdict is sound. Where separately-binding floors matter, exact certification requires an observation that separates every admissible pair on opposite sides of a floor (per-floor measurement is one sufficient design), a formal complement to the thesis, argued informally since Martinez-Alier, Munda, and O’Neill (1998), that weakly comparable values cannot be fused into one metric without loss. The requirement is sharpest where the floor constrains the productive base rather than its output: an aggregate of services can read at full level while the base regenerating them is reduced beneath it, and only an observation reaching the base itself can detect the discrepancy.

Bias. Remark 5 shows that the loss need not be in the measurement: a biased indicator with an uncorrected certainty-equivalence policy empties the epistemic kernel of a system whose perfect-information kernel is nonempty. Monitoring therefore has two obligations — bounding the observation error and supplying the correction — and the second lies entirely at the governance structure’s discretion: the indicator is published, the correction is not applied, and the kernel empties although the information was sufficient.

Institutions. The hierarchy of Section 2.1 records that epistemic contraction is one of three causes of kernel shrinkage, the others being disturbances and institutional restriction (authority, enforcement, allocation). The epistemic-institutional kernel $\text{IRViab}_{\mathcal{T}}(\mathcal{V})$ (Section 2.1) combines them: an institution restricted in what it may observe *and* in what it may command inherits both contractions. The obstruction calculus characterizes the first; its certificates remain valid, and typically tighten, when the institution’s command set is further restricted, since restriction of the action correspondence can only empty kernels, never create them (one-sided monotonicity of viability under correspondence restriction).

Proposition 8 (monotonicity of the epistemic kernel). The robust epistemic kernel is monotone in its constituents: enlarging the action sets, reducing the disturbance sets, refining the information structure, or enlarging the policy class never shrinks the kernel. Writing $\text{ERViab}^{\cdot}$ for the kernel with the indicated constituent, $U_1 \subseteq U_2$, $D_1 \subseteq D_2$, $\Pi_1 \subseteq \Pi_2$, and “ $\mathcal{I}_1$ refines $\mathcal{I}_2$ ” imply respectively

$$\text{ERViab}^{U_1} \subseteq \text{ERViab}^{U_2}, \quad \text{ERViab}^{D_2} \subseteq \text{ERViab}^{D_1},$$

$$\text{ERViab}^{\Pi_1} \subseteq \text{ERViab}^{\Pi_2}, \quad \text{ERViab}^{\mathcal{I}_2} \subseteq \text{ERViab}^{\mathcal{I}_1}.$$

Proof sketch. A policy viable under the smaller action sets, larger disturbance sets, smaller policy class, or coarser information structure remains admissible and nonanticipative under the corresponding enlargement or refinement, while the

adversary’s options are unchanged or reduced; the same policy witnesses viability.

6.5 Limitations

The certificates are sound but not complete; several extensions are noted as open.

- (i) The continuous-time certificates of Sections 3.2–3.3 and 3.5–3.7 are sufficient conditions for nonviability, not necessary-and-sufficient characterizations of the epistemic kernel (backward recursion is complete in the finite-horizon finite-system setting, Section 3.1); the middle ground (neither a Veliov-type condition nor an obstruction certificate) is open. In continuous settings completeness would additionally require a measurable-selection theory beyond the explicit selections the certificates exhibit; completeness is therefore claimed only where backward recursion is exact (Section 3.1).
- (ii) The timing obstruction of Theorem 3 (and its threshold form, Remark 2) certifies failure by exit *before* the first informative observation. It is blind to the complementary mode of nonviability — *insufficient post-observation recourse* — in which a common blind-window control keeps every compatible branch safe, yet no observation-adapted control can recover every branch once the distinguishing observation arrives; the finite-horizon recursion (Section 3.1) captures that mode exactly, and the oracle-recourse certificate of Section 3.7 certifies it in continuous time — soundly against every measurable information-adapted policy under an oracle relaxation of the post-observation class, with a worked three-branch instance whose delay threshold is exact; what remains open is a complete continuous-time characterization that does not relax the post-observation class. A discrete instance in which every one-step certificate is silent is given in the Supplementary Material (S3, A.3).
- (iii) The information-set semantics is set-membership; probabilistic or belief-space formulations require separate statements, although the common-action obstruction transfers mutatis mutandis to any semantics in which the policy must choose a single action per information state.
- (iv) The results are finite-dimensional and time-invariant; the delay-free setting makes the timing bound of Theorem 3 sharp, and delay systems require the retarded extension of the Dini comparison argument (Hale and Verduyn Lunel, 1993).
- (v) Theorem 5’s adversarial-realization step presupposes both the closed-graph regularity of Section 2.1 and the closed-loop existence hypothesis (H5.2) — D lower semicontinu-

ous or constant, or the convexified reading; without these the certificate remains a heuristic.

- (vi) The institutional consequences of Section 6.4 are design conclusions from the certificates, not empirical findings; their empirical testing belongs to applied studies.

7. Complete Characterizations and the Residual Gap

The certificates of Section 3 are sound sufficient conditions for nonviability, and Section 6.5 recorded that they do not exhaust the complement of the epistemic kernel. This section makes that gap precise: it exhibits two settings in which the viability question admits an exact answer — the one-step (finite discrete) setting, where the common-action certificate is necessary and sufficient, and the static-observation setting, where the problem reduces exactly to open-loop robust viability — and states the residual dynamic gap as an open problem.

Theorem 6 (one-step completeness). Let X, A, D, Y be finite, with transition $x^+ = F(x, a, d)$, observation $y = O(x, a, x^+)$, and safe set $\mathcal{V} \subseteq X$. Define the one-step safe set

$$S_1(x) = \{a \in U(x) : F(x, a, d) \in \mathcal{V} \text{ for all } d \in D(x)\}.$$

A belief $B \subseteq \mathcal{V}$ is one-step viable under $\mathcal{I}$ — some observation-based policy achieves $x_1 \in \mathcal{V}$ for every $x_0 \in B$ and every admissible disturbance — if and only if $\bigcap_{x \in B} S_1(x) \neq \emptyset$. Hence $B \notin \mathcal{W}_1$ exactly when the common safe-action set is empty, the discrete reading of Theorem 2.

Proof. ($\Rightarrow$) An observation-based policy depends on the record only, and every $x \in B$ produced the same record, so the policy plays a single $a \in U^B(B)$; safety of the first step forces $F(x, a, d) \in \mathcal{V}$ for all $x \in B$ and $d \in D(x)$, i.e. $a \in \bigcap_{x \in B} S_1(x)$. ($\Leftarrow$) Playing any a in the intersection keeps every successor in $\mathcal{V}$. $\square$

Combined with the backward recursion $\mathcal{W}_{k+1} = \text{Pre}(\mathcal{W}_k)$ of Theorem 1, Theorem 6 shows that the pair (*common-action certificate, recursion*) is a complete characterization at every finite horizon in finite systems: the certificate detects first-step emptiness, and the recursion propagates the obstruction to every later step.

Theorem 7 (static-observation reduction to open-loop viability). Let $\mathcal{V}$ be compact with C^1 constraint functions q_j, f continuous with D compact-valued and of closed graph, $U(x) \equiv U$, and suppose the observation is *static*: $y(t) = O(x(t))$ is constant along every admissible trajectory, so no information accrues over time. Then every observation-based policy coincides with an *open-loop* control — a measurable function of time only, $u(\cdot) : [0, \infty) \rightarrow U$ — and $B_0 \in \text{ERViab}_{\mathcal{I}}(\mathcal{V})$ if and only if there exists an open-loop control under which every trajectory from every $x_0 \in B_0$ and every admissible disturbance realization stays in $\mathcal{V}$ for all time. Within the constant-control subclass $u(\cdot) \equiv u_0$, the condition is the robust Nagumo boundary condition

$$\nabla q_j(x) \cdot f(x, u_0, d) \geq 0 \quad \text{for all } x \in \partial \mathcal{V}, \text{ active } j, d \in D(x),$$

necessary for a constant control to keep $\mathcal{V}$ invariant and sufficient under the standard regularity assumptions (Aubin, 1991; Frankowska, 1989). A time-varying open-loop control may be viable when no constant control is; the open-loop reduction, not the constant-control condition, is the exact characterization in the static class.

Proof. A static observation makes the record constant, so a policy’s action at time t can depend on time only — open-loop; this gives the reduction in both directions (a viable policy

yields a viable open-loop control, and every open-loop control is an admissible observation-based policy). The constant-control statement is the robust Nagumo theorem (Aubin, 1991; Frankowska, 1989): *necessary*, because any trajectory leaving $\mathcal{V}$ under u_0 exhibits a boundary point with outward drift, and *sufficient* under the regularity assumptions. $\square$

Theorem 7 makes the sense of completeness in the static class precise: the obstruction is exactly the nonexistence of a robustly viable open-loop control, and within the constant-control subclass the boundary condition of Theorem 2 is necessary and sufficient. Completeness fails only when information accrues, where the residual modes are post-observation recourse and timing — the hidden-mode example and the timing bound of Section 3.3.

The static-observation theorem localizes where completeness can fail, and the failure admits a precise structure. Consider the *two-phase model*: a window of K blind steps followed by exact branch revelation and branch-adapted continuation, with declared blind class Π_B and full-information kernel $\text{RViab}(\mathcal{V})$ available after the reveal.

Proposition 9 (exact two-phase decomposition). In the two-phase model, B_0 is viable if and only if (i) some declared blind control keeps every branch in $\mathcal{V}$ through step K , and (ii) some declared blind control satisfying (i) lands every branch in $\text{RViab}(\mathcal{V})$ at the reveal. Consequently, over any declared class on which the window certificates are complete — finite classes; polytope-declared classes, by Theorem 4 — the following are equivalent: both certificates are silent; clause (i) holds; and either B_0 is viable or its nonviability is exactly the post-observation recourse mode, the failure of (ii) for every window-surviving blind control.

Proof sketch. A policy for the two-phase model is a blind part followed by an adapted part. After the reveal the information is complete, so an adapted continuation exists from a revealed state if and only if that state lies in $\text{RViab}(\mathcal{V})$; and the blind part succeeds only branchwise, since a branch leaving $\mathcal{V}$ before the reveal, or landing outside the kernel at it, kills viability upon that branch’s realization. This gives the equivalence. Completeness of the window certificates turns silence into clause (i), and the residual failure of clause (ii) is the recourse mode; the complete proof is in Supplementary S1.

Proposition 10 (no window-measurable pair is complete). Call a certificate pair *window-measurable* when its two verdicts depend only on the window sub-model: the initial belief, the branch transitions within the window, the floors, and the observation schedule. Then no window-measurable pair is complete for the two-phase model: the three-state instance of Supplementary S3/A.3 and its variant agree on all window data, the certificates are silent in both, and yet one is nonviable (x_4 exits under every post-reveal action) while the other is viable (x_4 is maintained).

Proof sketch. The two instances differ only in x_4 ’s post-reveal transitions, so any window-measurable pair returns identical verdicts — here, silence on both. The exact recursion decides differently: in the first instance every post-reveal action exits $\mathcal{V}$, so $x_4 \notin \text{RViab}(\mathcal{V})$ and the belief misses $\mathcal{W}_2$; in the second x_4 is maintained, lies in the kernel, and the belief is viable. Both computations are machine-verified in exact rational arithmetic (Supplementary S1, verification record). The residual dynamic gap of Open Problem 1 is thereby narrowed to a definite target: any complete certificate pair for the dynamic class must read data beyond the window sub-model.

Open Problem 1 (dynamic-observation completeness). Give a finite, checkable certificate that is necessary and sufficient for membership in $\text{ERViab}_{\mathcal{I}}(\mathcal{V})$ under dynamic observation beyond the belief-space restatement below, or prove a separation

showing that none exists in a natural class. The belief-space restatement is: $B_0 \in \text{ERViab}_{\mathcal{I}}(\mathcal{V})$ if and only if B_0 belongs to the viability kernel of the set-valued belief dynamics under the observation constraint (Cardaliaguet, Quincampoix, and Saint-Pierre, 2007); this is a complete characterization on the infinite-dimensional space of compact beliefs, and the obstruction calculus of Section 3 is its finite, certifiable layer.

8. The Selector Principle, Dual Certificates, and Monitoring Adequacy

The certificates of Section 3 are stated mechanism by mechanism. This section records the one object they consolidate, the exactness class in which its emptiness is also complete, and the design converse; it adds no new certificate and modifies none.

The selector. For a declared policy class, information structure, review length Δ , and horizon N , the recursively viable selector at information state B is

$$\Gamma_N(B) = \{\alpha : \text{Reach}_{[0, \Delta]}(B, \alpha) \subseteq \mathcal{V} \text{ and every successor belief } B' \in \mathcal{W}_{N-1}\},$$

with $\mathcal{W}_0 = \{B \subseteq \mathcal{V}\}$ and $\mathcal{W}_N = \{B : \Gamma_N(B) \neq \emptyset\}$ — the recursion of Theorem 1 — whose one-step factors are the ladder of Proposition 3. The principle: viability within a declared class is nonemptiness of this selector, and each certificate of Section 3 is a checkable witness that one factor of it is empty.

Dual witnesses. The witnesses, by factor: the common-action factor $\mathcal{R}_{\mathcal{V}}^B(B) = \emptyset$, witnessed by the Farkas pair of Section 3.2 and, under convexity, by the $m + 1$ -point sparse witness of Proposition 4; the tube factor, witnessed by the uniform-margin review obstruction of Proposition 2; the timing factor, witnessed by the threshold form $\sigma^*(B_0) < T_{\text{obs}}$ of Remark 2 and Theorem 3; finite-time exit, witnessed by the outward drift of Theorem 5 already at singletons; epistemic emptiness, witnessed by the minimal admissibility construction of Proposition 5; certification failure, witnessed by a fibre meeting both K and K^c , with the certainly-safe set of Corollary 1 (Proposition 7); and class restriction, witnessed by the certainty-equivalence trap of Remark 5 — the uncorrected law’s strictly positive drift against the corrected law’s nonpositive drift.

Exactness class. Weak duality holds unconditionally: a dual witness certifies nonviability in the declared class. Strong duality — emptiness if and only if a witness exists — holds exactly in the classes of Theorem 6 (finite horizon, finite model), Theorem 7 (static observation), and the delayed class of Remark 6; beyond them the converse is Open Problem 1, open. The causes are not unified: dynamic exit empties the selector under full information and is not epistemic; the fibre results are limits on certification, not on policy viability (Section 4); the trap is a class restriction and can fire under injective observation. Unifying the form while keeping the causes distinct is the strongest unifying statement the framework supports, and the companion unification statement carries the same discipline.

Monitoring adequacy (the design converse). Obstruction answers why nonviability is forced; design asks what must be measured. In the one-step exact class the converse is immediate: an observation structure is adequate on a target exactly when every fibre meeting the target has a nonempty common safe action — the partition form of the fibre criterion (Proposition 7); pairwise intersection is necessary and, off convexity, not sufficient (the three two-point codices of the companion monitoring paper). Four rules follow as identities

on the audited systems. (i) *Timing*: on the delayed hidden-regime hold class, viability at deadline T_{obs} holds exactly when $T_{\text{obs}} \leq \sigma^*(B_0) = z_0 - 1$, equivalently $z_0 \geq 1 + T_{\text{obs}}$; the boundary case $z_0 = 1 + T_{\text{obs}}$ is viable — the review must land no later than worst-case exit (Section’s grid; the strict form fails at the boundary cell, as the verification script records). (ii) *Coarseness*: a fibre merging states with disjoint safe-action sets obstructs at that information state while every branch remains viable (Theorem 2); the remedy is a fibre split, not a finer threshold on the same index. (iii) *Aggregation*: an index whose fibres cross a floor cannot support an exact safe/unsafe verdict; report the certainly-safe set or fall silent (Corollary 1). (iv) *Bias*: the trap of Remark 5 empties a nonempty kernel under the uncorrected law; the remedy is structural correction of the observation map, not a better sensor. The consolidated statement, with every audited count cited to its system, is maintained in the supplementary material.

9. A Calibrated Case Study and a Coverage Audit

We instantiate the three observation-side mechanisms on a concrete resource model and audit the certificates against the exact recursion of Theorem 1. The model is a single-species stock $S \in [0, 1]$ (fraction of carrying capacity) with discrete dynamics

$$S^+ = \text{clip}(S + S(1 - S) - u + w), \\ u \in [0, 0.3], \quad w \in \{-0.02, 0, +0.02\},$$

harvest u , recruitment noise w read adversarially, and a sustainability floor $S_{\min} = 0.3$ (Clark, 1990).

The fibre criterion. The regulator observes a coarse two-bin index, low if $S < 0.6$ and high if $S \geq 0.6$. The safe set $\mathcal{V} = [0.3, 1]$ is not a union of index bins: the low bin contains both 0.1 (below the floor) and 0.4 (above it). By Proposition 7 no exact observation-only certifier exists, and by Corollary 1 the certainly-safe set is the single bin $\mathcal{V}_{\text{safe}} = \{\text{high}\}$. A verdict of “safe” on the strength of a low reading is therefore unsound; the index can certify safety only when it reads high.

Two-dimensional aggregation. The fibre criterion also governs a multidimensional stock. Take a two-species state $(S_1, S_2) \in [0, 1]^2$ with the floor $S_1 \geq 0.4$, and suppose the regulator observes only the aggregate $I = S_1 + S_2$. The states $(0.5, 0.5)$ and $(0.2, 0.8)$ are both compatible with the reading $I = 1.0$, yet the first satisfies the floor and the second violates it, so by Proposition 7 no exact observation-only certifier exists; by Corollary 1 the certainly-safe readings are exactly $I \geq 1.4$ (the smallest S_1 compatible with I is $I - 1$, so $I - 1 \geq 0.4$). An aggregate reading below 1.4 must therefore fall silent, however carefully thresholded — the two-dimensional form of the aggregation consequence of Section 6.4.

The delayed-information timing bound. The timing obstruction is sharpest in the hidden-regime system of Section 3.3: regimes $\theta \in \{-1, +1\}$, dynamics $z^+ = z + \theta u$, controls $u \in \{-1, +1\}$, floor $z \geq 1$, with θ revealed only after a blind window of T_{obs} steps. From $z_0 = 2$ the drift of $q(z) = z - 1$ on the declining branch is -1 under every admissible action, so the threshold of Theorem 3 and Remark 2 is $\sigma^*(B_0) = z_0 - 1 = 1$: nonviability is certified for every $T_{\text{obs}} > 1$, while $T_{\text{obs}} = 1$ is viable — the regime is revealed in time, and $u = \theta$ then restores growth. In the fishery above the same certificate is silent in the normal regime, because $u_0 = 0$ keeps every stock above the floor; the bound fires only when no blind action avoids decline, which is precisely its scope.

The certainty-equivalence trap. With a biased index $\hat{S} = S + 0.1$, $g(S) = S^2$, and $\mathcal{V} = [1, 2]$, the certainty-equivalence policy $u = g(\hat{S})$ produces $\dot{S} = g(S + 0.1) - g(S) = 0.2S + 0.01 > 0$, which reaches the upper bound from $S = 1$ in

≈ 3.3 time units, while the bias-corrected policy $u = g(\hat{S} - 0.1)$ keeps $\hat{S} = 0$ and the kernel nonempty (Remark 5).

A reproducible delayed hidden-regime model. The audit runs on the exact instance specified here. The state is (z, θ) with $\theta \in \{-1, +1\}$ a hidden regime and $z \geq 0$; the action $u \in \{-1, +1\}$ is common to both regimes; the transition is $z^+ = z + \theta u$; the safe set is $\mathcal{V} = \{z \geq 1\}$; and the observation reveals nothing before T_{obs} and reveals θ exactly at time T_{obs} . From a known z_0 with both regimes possible, the belief after $k < T_{\text{obs}}$ blind steps contains the two extreme histories $(z_0 - k, -1)$ and $(z_0 + k, +1)$, and a blind policy must choose the same u for both branches at every blind step. The guaranteed blind-window survival time is therefore $\sigma^*(B_0) = z_0 - 1$: the best blind policy drives one branch down at unit rate, so the declining branch reaches the floor 1 at time $z_0 - 1$, and after the reveal $u = \theta$ keeps $z \geq 1$ forever. Hence B_0 is robustly epistemically viable if and only if $z_0 - 1 \geq T_{\text{obs}}$, i.e. $z_0 \geq 1 + T_{\text{obs}}$ — the ground truth against which the certificates are audited. The one-step certificate of Theorem 2 fires exactly when $z_0 < 2$ (one blind step already forces $z_0 - 1 < 1$), and the timing bound of Theorem 3 fires exactly when $z_0 \geq 2$ and $T_{\text{obs}} > z_0 - 1$.

A coverage audit. To quantify the gap identified in Section 6.5, we audit the certificates against the exact recursion on the delayed hidden-regime system, over the grid $z_0 \in \{1.0, 1.1, \dots, 2.5\}$, $T_{\text{obs}} \in \{1, 2, 3\}$. Table 2 and Figure 3 report the true verdict, the one-step certificate, and the timing bound. The audit is reproduced by the script `paper2_coverage_audit.py`, publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`), which regenerates Table 2 and Figure 3 verbatim.

Remark 6 (joint completeness in the delayed class). In every audited cell the true verdict is reproduced by exactly one of the two certificates: of the 42 nonviable cells, 30 are certified by the common-action certificate and the remaining 12 — precisely those with $z_0 \geq 2$ and $T_{\text{obs}} > z_0 - 1$ — by the timing bound; none is uncovered. In this class the two certificates are jointly complete, and the Section 6.5 gap reduces to the timing cells.

Why joint completeness holds in this class. The ground truth is the identity $B_0 \in \text{ERViab}_{\mathcal{I}}(\mathcal{V}) \iff z_0 \geq 1 + T_{\text{obs}}$ established in the reproducible-model paragraph above, so nonviability holds exactly when $z_0 < 1 + T_{\text{obs}}$. The two certificates partition that set: the common-action certificate fires exactly when $z_0 < 2$, and the timing bound fires exactly when $z_0 \geq 2$ and $T_{\text{obs}} > z_0 - 1$, that is $2 \leq z_0 < 1 + T_{\text{obs}}$; the two conditions are disjoint and their union is $\{z_0 < 1 + T_{\text{obs}}\}$. Every nonviable cell is therefore certified by exactly one certificate, so within the delayed hidden-regime class the pair (common-action, timing) is a complete characterization at every horizon, and the gap of Section 6.5 is exactly the timing cells.

Calibration. The certificate parameters correspond to quantities estimable from standard stock-assessment practice: the drift margin ε is a net growth rate at the floor; the first informative observation time T_{obs} is the assessment or review periodicity; the bias b of the certainty-equivalence instance is the systematic error of the indicator; and the floor is the reference point of the management rule. Calibrating these four numbers from a published assessment is all the machinery the certificates require; the present section stays symbolic, and the empirical calibration is left to applied studies.

A two-patch protection audit. The aggregation obstruction is not a one-dimensional phenomenon, and the certificates read it directly in two dimensions. Two patches $i \in \{1, 2\}$ carry stocks $z_i \in \{0, 1, 2, 3\}$, each with recruitment $+1$ per step and a protection resource that covers one patch per step: the action $u \in \{1, 2\}$ protects patch u , the protected patch loses nothing, and the other loses two units, $z_i^+ = \min\{3, z_i + 1 - 2 \cdot \mathbf{1}[u \neq i]\}$.

The sustainability floor is $z_i \geq 1$ for both patches, and the regulator observes only the aggregate reading $y = z_1 + z_2$. From the reading $y = 3$ the compatible safe states are $(1, 2)$ and $(2, 1)$; the first is safe only under $u = 1$ and the second only under $u = 2$, so the common safe-action set is empty and the common-action certificate (Theorem 2) fires — while each state alone lies in the full-information kernel (protect the low patch, then alternate), the minimal epistemic-emptiness form of Section 3.4 realized in two dimensions. The certainly-safe readings are exactly $y \geq 4$: every state compatible with $y \leq 3$ includes one with a depleted patch, so the aggregated index can certify safety only above that threshold — the two-dimensional reading of Corollary 1. Table 3 audits the aggregated beliefs against the exact belief-level recursion, in which the reading splits the belief into its y -cells at every step: every nonviable belief is certified, and all by the common-action certificate — the aggregation obstruction is instantaneous, and no observation schedule rescues it, in contrast with the timing cells of the one-dimensional audit above, where $T_{\text{obs}} = 1$ is viable. The full-information kernel is $\text{RViab}(\mathcal{V}) = \{z_1, z_2 \geq 1\} \setminus \{(1, 1)\}$: the double-depleted state is nonviable outright, one protection per step cannot hold both floors — an instance of nonviability beneath all epistemic analysis. Both computations are machine-verified in exact integer arithmetic, and the verification script regenerates the table verbatim.

The audit is symbolic and one-dimensional, with two two-dimensional instances (the aggregated reading above and the two-patch protection audit); a full multidimensional belief-space numerical campaign is deferred to future work, and the case study makes no claim of a general-purpose computational calculus.

10. A Probabilistic Development: Belief-State Safety Values

The certificates of Sections 3–4 rest on set-membership semantics: the information state is a set of compatible states. This section records the stochastic counterpart — the belief-state safety value of partially observable control — and its obstruction certificate, connecting the calculus to the partially observable literature (Åström, 1965; Smallwood and Sondik, 1973).

Let X, A, D, Y be finite, with a stochastic transition $T(x' | x, a)$ and an observation likelihood $g(y | x, a, x')$. Augment the state space with an absorbing unsafe state $\perp$: for $x \in \mathcal{V}$, $T(\perp | x, a) = 1 - \sum_{x' \in \mathcal{V}} T(x' | x, a)$, and $T(\perp | \perp, a) = 1$. A belief is a distribution b over $X \cup \{\perp\}$, updated by Bayes' rule $\tau(b, a, y)$, with $\mathbb{P}(y | b, a) = \sum_{x, x'} b(x) T(x' | x, a) g(y | x, a, x')$. The *safety value* is the maximal survival probability

$$V_k(b) = \max_{\pi} \mathbb{P}_{\pi}(x_t \in \mathcal{V} \text{ for } t = 0, \dots, k | b).$$

Theorem 8 (belief-state safety value). $V_0(b) = b(\mathcal{V})$ (the horizon counts the current state), and for $k \geq 0$

$$V_{k+1}(b) = \max_{a \in A} \sum_{y \in Y} \mathbb{P}(y | b, a) V_k(\tau(b, a, y)),$$

with the time convention that the policy acts, observes y , then acts again. The recursion is exact on every finite horizon; it is the Smallwood–Sondik belief-state value iteration specialized to the absorbing safety reward (Smallwood and Sondik, 1973).

Proof. Standard value iteration for the belief-state Markov decision process: conditioning on the observation after the action and applying the optimal continuation gives the identity, and the absorbing state makes V_k the k -step survival probability (Smallwood and Sondik, 1973). $\square$

T_{obs}	initial position z_0																
	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0	2.1	2.2	2.3	2.4	2.5	
1	●	●	●	●	●	●	●	●	●	●	○	○	○	○	○	○	
2	●	●	●	●	●	●	●	●	●	●	*	*	*	*	*	*	
3	●	●	●	●	●	●	●	●	●	●	*	*	*	*	*	*	

Table 2: Coverage audit on the delayed hidden-regime system: ○ viable; • nonviable, certified by the common-action certificate (Theorem 2); * nonviable, certified only by the timing bound (Theorem 3). Every nonviable cell is covered by exactly one certificate.

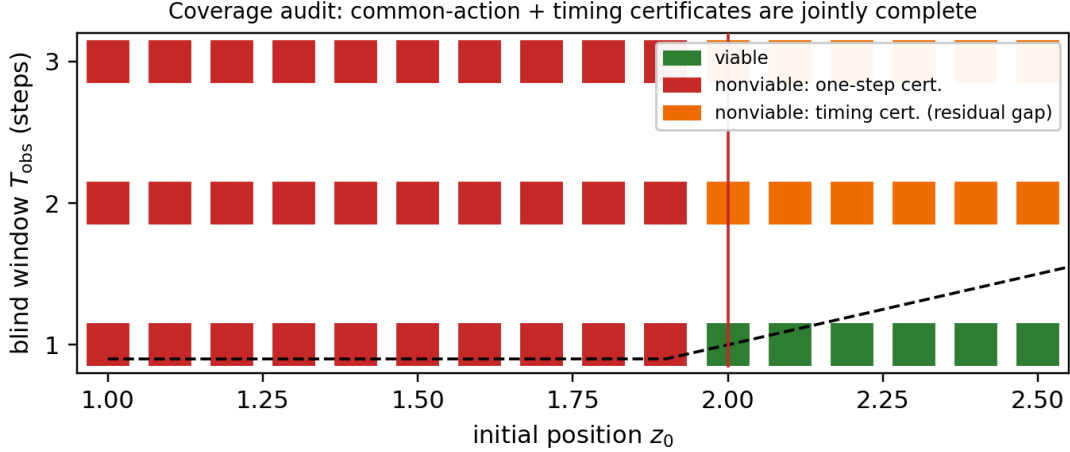


Figure 3: The coverage audit of Table 2: green = viable; red = nonviable, certified by the common-action certificate; orange = nonviable, certified only by the timing bound (the residual gap of Section 6.5). The dashed curve is the true boundary $z_0 = 1 + T_{\text{obs}}$ and the solid vertical line the one-step boundary $z_0 = 2$.

initial belief	reading y	verdict	covering certificate / witness
(2, 2)	4	○ viable	witness: $u = (1, 2)$ cycling
(1, 2), (2, 2)	3, 4	○ viable	witness: $u = (1, 2)$ cycling
(2, 1), (2, 2)	3, 4	○ viable	witness: $u = (2, 1)$ cycling
(1, 2), (2, 1)	3	•	common-action (Theorem 2)
(1, 2), (2, 1), (2, 2)	3, 4	•	common-action (Theorem 2)
(1, 1)	2	•	common-action (Theorem 2)

Table 3: The two-patch protection audit: aggregated beliefs under the reading $y = z_1 + z_2$, audited against the exact belief-level recursion. Every nonviable belief is certified by the common-action certificate (Theorem 2); the viable beliefs carry explicit blind witness cycles.

Proposition 11 (chance-constrained common-action obstruction). Let $b \in \Delta(\mathcal{V})$ (so $b(\mathcal{V}) = 1$) and write $p(x, a) = 1 - \sum_{x' \in \mathcal{V}} T(x' | x, a)$. If

$$\sum_{x \in \mathcal{V}} b(x) p(x, a) \geq \delta \quad \text{for every } a \in A,$$

then $V_k(b) \leq 1 - \delta$ for every $k \geq 1$, and the chance constraint $\mathbb{P}_{b, \pi}(x_0, \dots, x_k \in \mathcal{V}) \geq 1 - \varepsilon$ is infeasible under every policy whenever $\varepsilon < \delta$.

Proof. The one-step survival probability under action a is $1 - \sum_x b(x) p(x, a) \leq 1 - \delta$, so $V_1(b) \leq 1 - \delta$. Since $V_{k+1} \leq V_k$ — more steps cannot raise the survival probability — $V_k(b) \leq V_1(b) \leq 1 - \delta$. □

Proposition 12 (degenerate (deterministic) limit). Let X, A, D, Y be finite with the deterministic maps $x^+ = F(x, a, d)$ and $y = O(x, a, x^+)$ of Section 3.1, and let the stochastic transition and likelihood have supports $\{F(x, a, d) : d \in D(x)\}$ and $\{O(x, a, x^+) : x^+ \text{ reachable}\}$. Then the posterior support under the belief-state update is exactly the set-valued post-state $\text{Post}(B, a, y)$ of Section 3.1, so the support of

the belief-state recursion follows the set-valued recursion $\mathcal{W}_k$. For a belief b carried by $B \subseteq \mathcal{V}$ and the disturbance read adversarially within its support, $V_k(b) = 1$ if $B \in \mathcal{W}_k$, while if $B \notin \mathcal{W}_k$ then $V_k(b) \leq 1 - \min_{x \in B} b(x) < 1$.

Proof. Under the deterministic kernels the posterior is supported exactly on $\text{Post}(B, a, y)$, so the support of the belief evolves by the set-valued recursion. If $B \in \mathcal{W}_k$, the witness actions of Theorem 1 keep every compatible branch in $\mathcal{V}$, giving $V_k(b) = 1$. If $B \notin \mathcal{W}_k$, every action leaves some compatible state $x \in B$ of mass $b(x)$ with a successor outside $\mathcal{V}$ under an admissible disturbance, so the one-step survival probability under every action is at most $1 - \min_{x \in B} b(x)$; since V_k is non-increasing in k , $V_k(b) \leq 1 - \min_{x \in B} b(x) < 1$. □

Proposition 12 is the sense in which the chance-constrained obstruction degenerates to the common-action obstruction: at the deterministic limit $V_1(b) = 1$ exactly when the common safe-action set is nonempty, and the deficit from 1 is at least the smallest compatible mass $\min_{x \in B} b(x)$. The bound is attained on the worked instances of the Supplementary Material (S1): the two-floor conflict at $V_k(b_0) = \frac{1}{2}$ for all k , and the delayed hidden regime on the audit grid of Section 9, where the script regenerates all 48 cells of the safety-value table verbatim.

11. Conclusion

Under standard finite-dimensional regularity assumptions, viability under perfect measurement has a mature kernel, tangency, and approximation theory. Viability under incomplete observation has long been asymmetric: the sufficiency side is carried by Veliov’s output-feedback condition and the estimation-tube programme, while the failure side has received less systematic treatment. This paper supplies the missing instrument: the obstruction calculus develops sound, computationally interpretable certificates for nonviability — that is,

sufficient conditions for nonviability, equivalently necessary conditions for viability. They do not exhaust the complement of the epistemic kernel (Section 6.5), and the paper claims no complete characterization. The mechanisms are organized by one selector principle (Proposition 1) — observation-based viability is the existence of a common recursively safe response — and are nested in an obstruction ladder (Proposition 3); in finite systems backward belief recursion is sound and complete (Theorem 1), so the calculus is exact where it can be and sound elsewhere.

With displayed proofs and explicit witnesses, the certificates identify five mechanisms by which an observation structure can make robust viability impossible, plus one through which a single uncorrected policy empties the kernel: the disturbance can enforce exit in finite time against every control; a constant observation can merge states with incompatible admissible controls (Proposition 5’s admissibility form); the compatible states can demand incompatible actions at a single information state; the information can arrive after the enforced exit; and the observation fibres can cross the safe-set boundary, defeating every exact certifier — with the uncorrected certainty-equivalence controller as the sixth, policy-specific, mechanism. Each certificate that admits a design remedy identifies it: a faster indicator, a separating observation, an observation that separates the sides of each floor instead of aggregate certification, or the correction of a known bias. For sustainability governance, where observation is conducted through coarse indicators at discrete reviews, the calculus states the quantitative limits of what monitoring can deliver — and converts each impossibility into a design specification. Four elements complete the picture: the certificates are complete in the one-step and static-observation classes, with the residual dynamic gap stated as an open problem (Section 7); the selector principle consolidates the certificates as dual witnesses of one correspondence, with the monitoring-adequacy converse and the design rules (Section 8); a calibrated case study and a coverage audit show the certificates reproducing the exact recursion verdict on a grid of delayed hidden-regime systems, with a two-patch audit carrying the aggregation obstruction to two dimensions (Section 9); and the probabilistic belief-state lift connects the calculus to the partially observable control literature, its safety values worked in closed form on the same grid (Section 10). A fourth completion is structural: over polytope-declared blind-window classes the timing certificate is exactly computable as a linear program, sound beyond the declaration with an unbounded implementable-class gap (Section 3.3), and the residual dynamic gap decomposes exactly, its post-observation recourse phase undecidable by any window-measurable certificate pair (Section 7).

References

- Åström, K.J.: Optimal control of Markov processes with incomplete state information. *J. Math. Anal. Appl.* **10**, 174–205 (1965)
- Abae, A.: Aggregate indices and transition safety: a quantifier-order separation between scalarized and coordinate-wise feasibility. <https://doi.org/10.6084/m9.figshare.33764023> (2026)
- Aubin, J.-P.: *Viability Theory*. Birkhäuser, Boston (1991)
- Aubin, J.-P.: Viability kernels and capture basins of sets under differential inclusions. *SIAM J. Control Optim.* **40**(3), 853–871 (2001)
- Aubin, J.-P., Bayen, A.M., Saint-Pierre, P.: *Viability Theory: New Directions*, 2nd edn. Birkhäuser, Boston (2011)
- Aubin, J.-P., Catté, F.: Bilateral fixed-points and algebraic properties of viability kernels and capture basins of sets. *Set-Valued Anal.* **10**, 379–416 (2002)
- Aubin, J.-P., Frankowska, H.: *Set-Valued Analysis*. Birkhäuser, Boston (1990)
- Béné, C., Doyen, L., Gabay, D.: A viability analysis for a bio-economic model. *Ecol. Econ.* **36**, 385–396 (2001)
- Cardaliaguet, P., Quincampoix, M., Saint-Pierre, P.: Differential

games through viability theory: old and recent results. In: Jørgensen, S., Quincampoix, M., Vincent, T.L. (eds.) *Advances in Dynamic Game Theory*. Annals of the International Society of Dynamic Games, vol. 9, pp. 3–35. Birkhäuser, Boston (2007)

- Clark, C.W.: *Mathematical Bioeconomics: The Optimal Management of Renewable Resources*, 2nd edn. Wiley, New York (1990)
- De Lara, M., Doyen, L.: *Sustainable Management of Natural Resources: Mathematical Models and Methods*. Springer, Berlin (2008)
- Doyen, L., Gajardo, P.: Sustainability standards, multicriteria maximin, and viability. *Nat. Resour. Model.* **33**(3), e12250 (2020)
- Doyen, L., Thébaud, O., Béné, C., Martinet, V., Gourguet, S., Bertignac, M., Fifas, S., Blanchard, F.: A stochastic viability approach to ecosystem-based fisheries management. *Ecol. Econ.* **75**, 32–42 (2012)
- Farkas, J.: Theorie der einfachen Ungleichungen. *J. Reine Angew. Math.* **124**, 1–27 (1902)
- Frankowska, H.: Optimal trajectories associated with a solution of contingent Hamilton–Jacobi equations. *Appl. Math. Optim.* **19**, 291–311 (1989)
- Gale, D.: *The Theory of Linear Economic Models*. McGraw-Hill, New York (1960)
- Haddad, G.: Monotone viable trajectories for functional-differential inclusions. *J. Differ. Equ.* **42**, 1–24 (1981)
- Hale, J.K., Verduyn Lunel, S.M.: *Introduction to Functional Differential Equations*. Springer, New York (1993)
- Kurzhanski, A.B., Vályi, I.: *Ellipsoidal Calculus for Estimation and Control*. Birkhäuser, Boston (1997)
- Maghenem, M., Sanfelice, R.G.: Characterizations of safety in hybrid inclusions via barrier functions. In: *Proceedings of the 22nd ACM International Conference on Hybrid Systems: Computation and Control (HSCC)*, pp. 109–118 (2019)
- Martinez-Alier, J., Munda, G., O’Neill, J.: Weak comparability of values as a foundation for ecological economics. *Ecol. Econ.* **26**(3), 277–286 (1998)
- Nagumo, M.: Über die Lage der Integralkurven gewöhnlicher Differentialgleichungen. *Proc. Phys.-Math. Soc. Japan* **24**, 551–559 (1942)
- Prajna, S., Jadbabaie, A.: Safety verification of hybrid systems using barrier certificates. In: Alur, R., Pappas, G.J. (eds.) *Hybrid Systems: Computation and Control (HSCC 2004)*. LNCS, vol. 2993, pp. 477–492. Springer, Berlin (2004)
- Prajna, S., Jadbabaie, A., Pappas, G.J.: A framework for worst-case and stochastic safety verification using barrier certificates. *IEEE Trans. Autom. Control* **52**(8), 1415–1428 (2007)
- Prajna, S., Rantzer, A.: On the necessity of barrier certificates. *IFAC Proc. Vol.* **38**(1), 526–531 (2005)
- Quincampoix, M., Veliov, V.M.: Viability with a target: theory and applications. In: *Control Theory, Multivariate Analysis and Applications*, pp. 47–63. Springer (1994)
- Saint-Pierre, P.: Approximation of the viability kernel. *Appl. Math. Optim.* **29**, 187–209 (1994)
- Smallwood, R.D., Sondik, E.J.: The optimal control of partially observable Markov processes over a finite horizon. *Oper. Res.* **21**(5), 1071–1088 (1973)
- Sontag, E.D.: *Mathematical Control Theory: Deterministic Finite Dimensional Systems*, 2nd edn. Springer, New York (1998)
- Veliov, V.M.: Sufficient conditions for viability under imperfect measurement. *Set-Valued Anal.* **1**, 305–317 (1993)
- Witsenhausen, H.S.: A counterexample in stochastic optimum control. *SIAM J. Control* **6**(1), 131–147 (1968)

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